

INCLUSIVE DIGITAL TECHNOLOGIES

WaySense

Find What You Need. With No Barriers.

FairTech

Asena Kucukali

Burce Karayazi

Vivek Kumar Sharma

OUR TEAM



Asena Kucukali
Management Engineering
11033222



Burce Karayazi
Management Engineering
11022387



Vivek Kumar Sharma
Compter Science Engineering
11063788

CONTENT

- Introduction
- Problem Definition
- User Needs
- Evaluation of Idea
- First Prototype
- Feedback Based Modifications
- Final Prototype
- Feasibility
- Future Work

INTRODUCTION

- Initial analysis focused on Zara's website and store experience
- Identified gaps in support for users with ADHD
- Too many product options
- Lack of guidance



PROBLEM DEFINITION

- ADHD affects ~5% of the global population

- Common challenges in shopping:

- **Cognitive overload** :especially in busy stores
- **Difficulty filtering distractions:** leading to forgotten goals
- **Too many choices:** leads to decision fatigue

- Result: Incomplete shopping journeys and exclusion

USER NEEDS

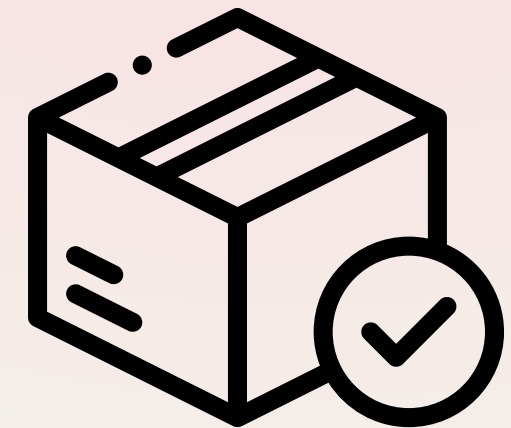
Guided Product Discovery

- In-store: route to item with navigating physical space
- Online: reduce complexity of filters etc. for finding relevant items



Focused Product Information

- Avoid overwhelming or vague descriptions
- Provide essential details like style, color, cut



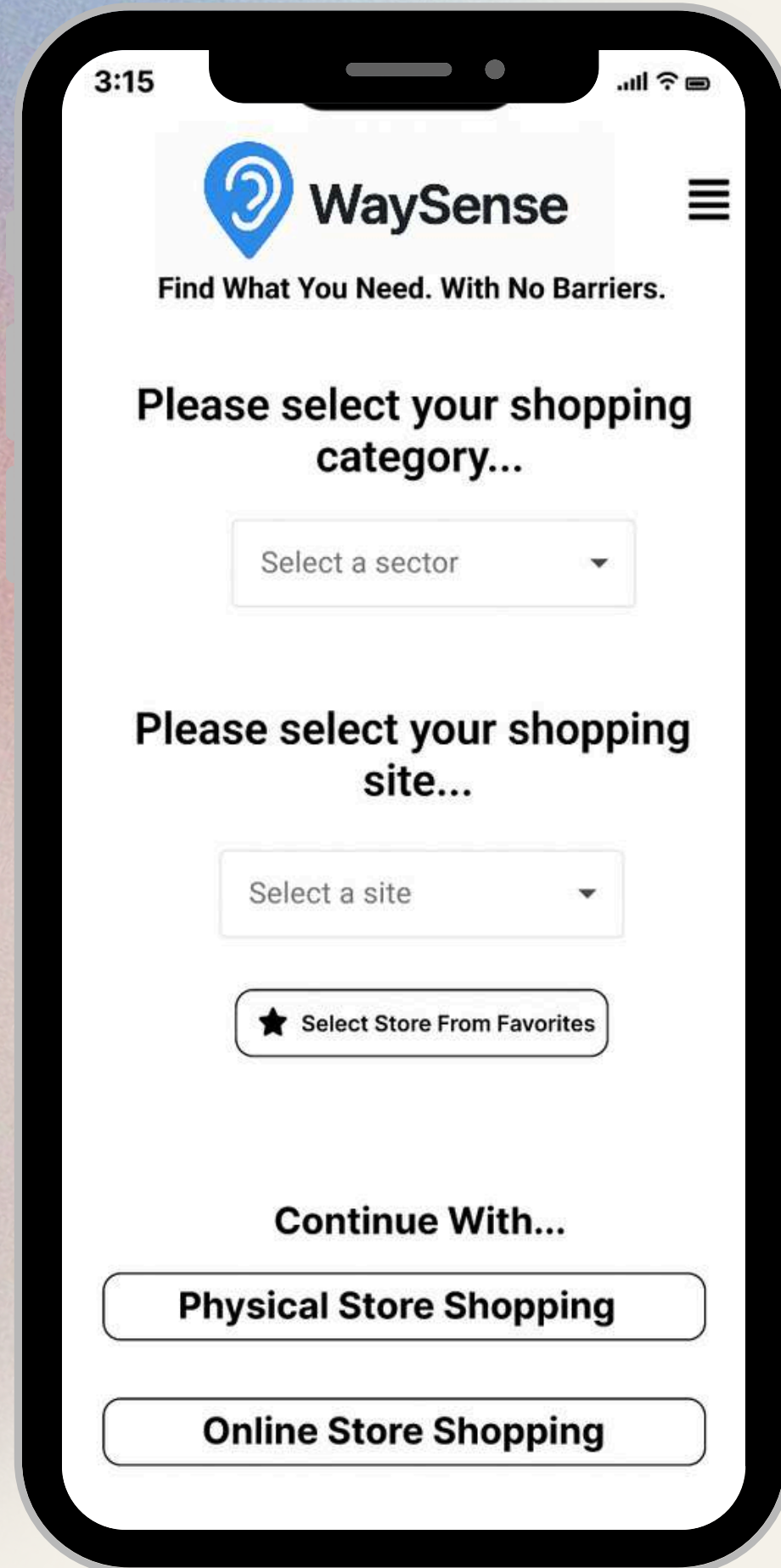
EVALUATION OF IDEA

- Indoor Navigation
- AI-Based Product Description Generator

Benefits:

- Works for both online and in-store settings
- Scalable and easy to integrate
- Aligned with inclusive design principles

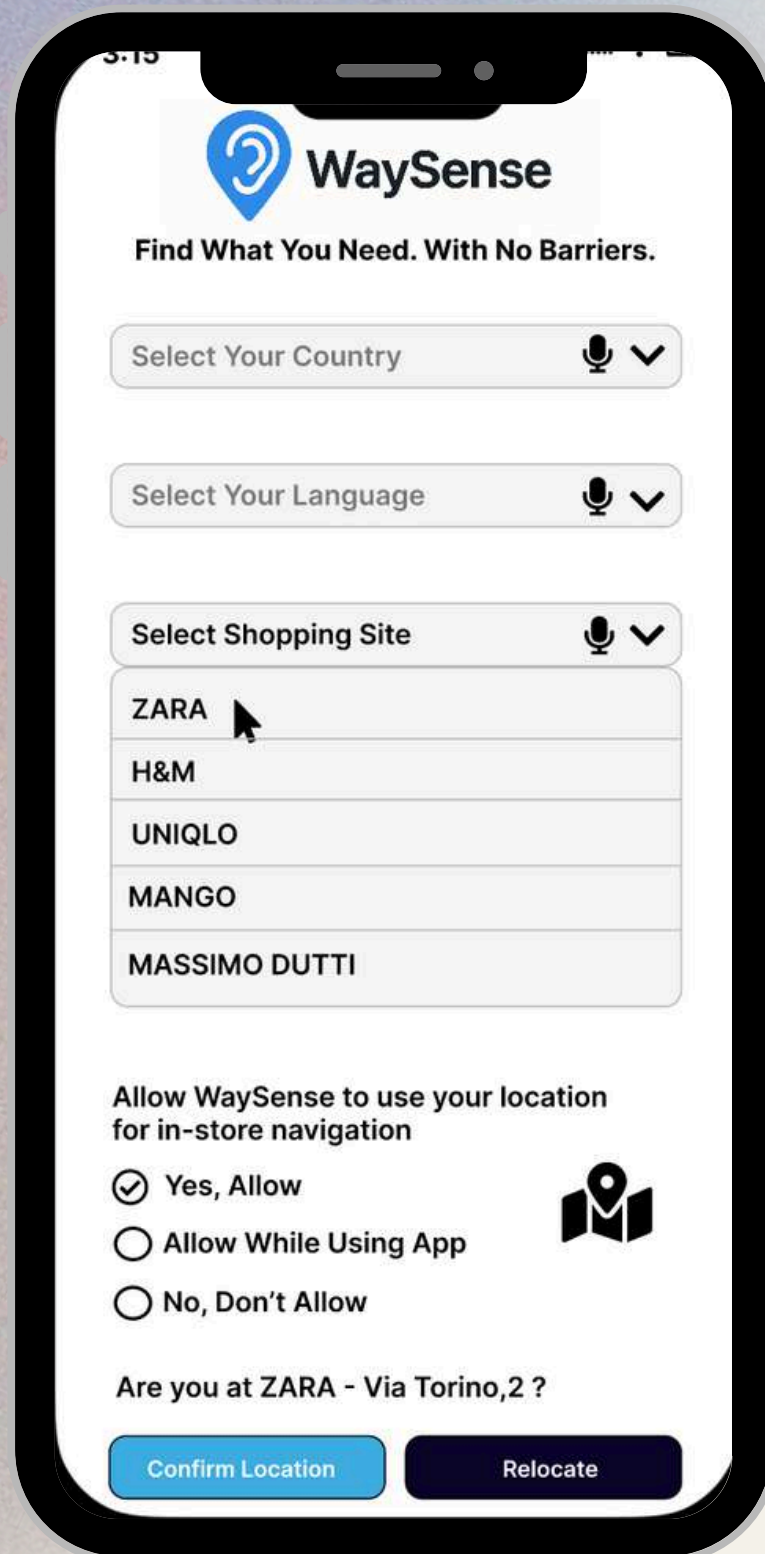




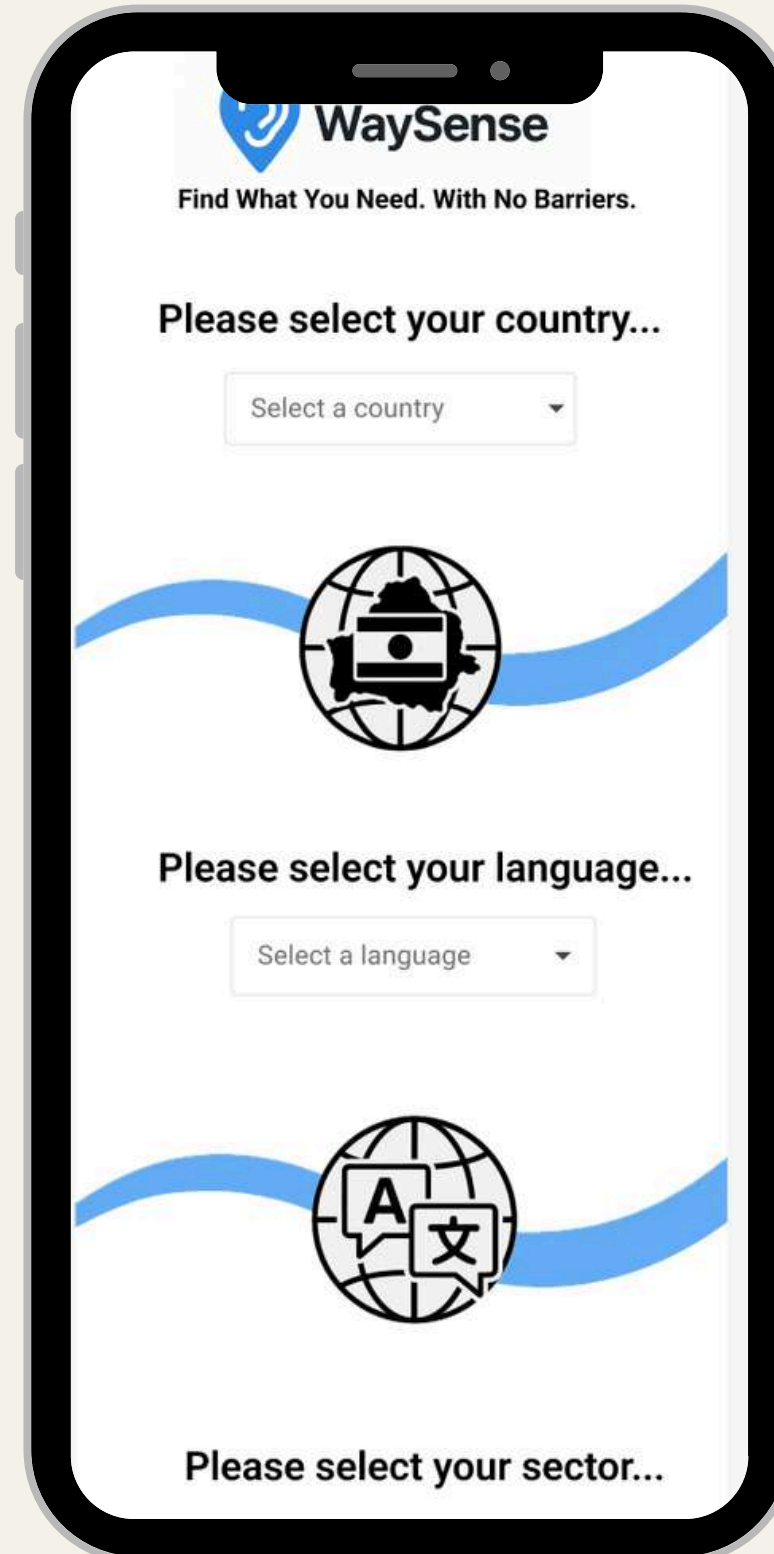
WAYSENSE

- A toolkit for retailers
- Smart product descriptions for clarity
- Indoor navigation to guide in physical stores
- Minimal UI, low decision load
- Flexible integration for clothing, hardware, supermarket etc. platforms

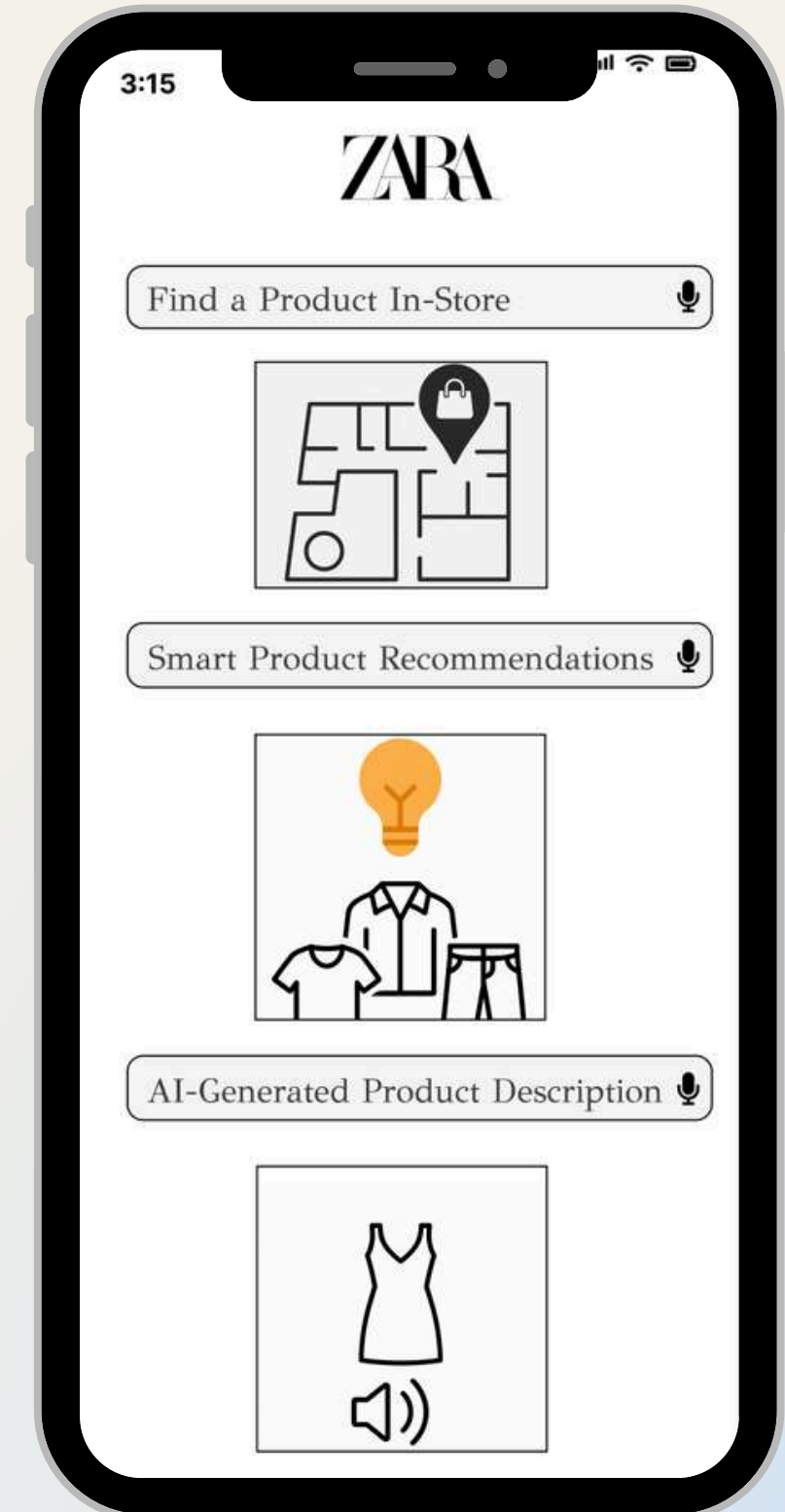
FIRST PROTOTYPE



No Sector Selection

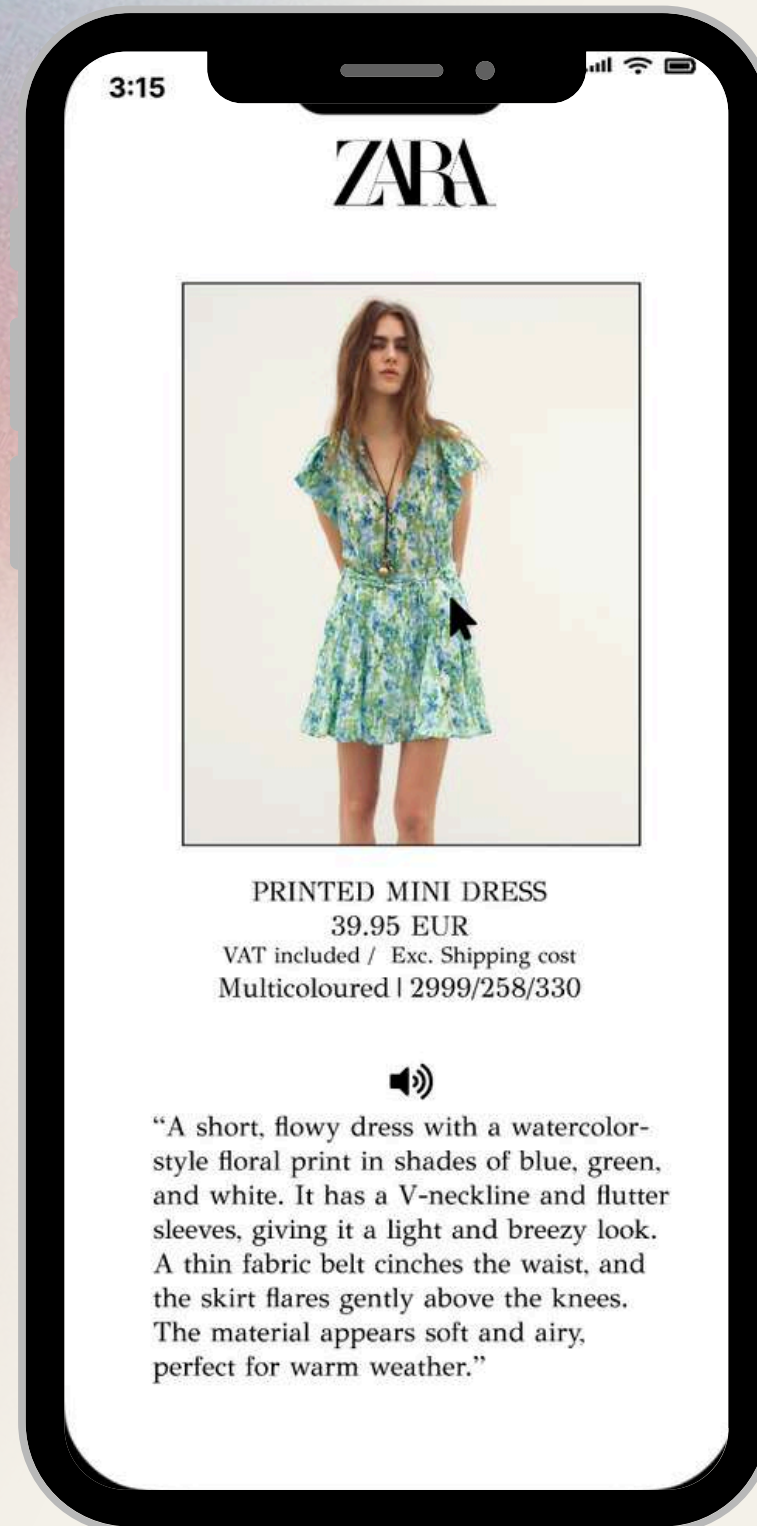


Too Many Questions

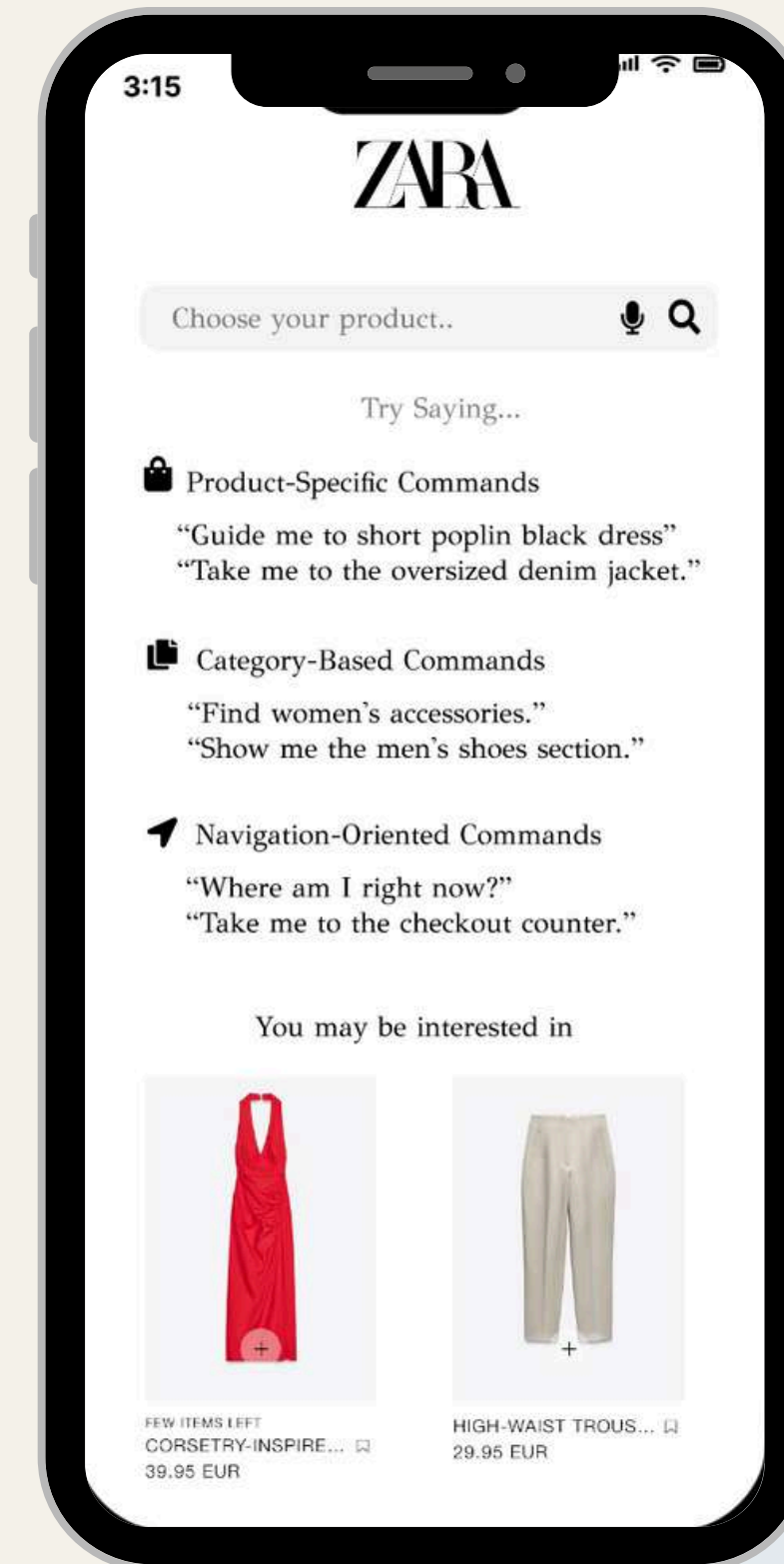


Too Many Features

FIRST PROTOTYPE



Long Descriptions without
highlighted words



No Home Page Button

FEEDBACK-BASED MODIFICATIONS

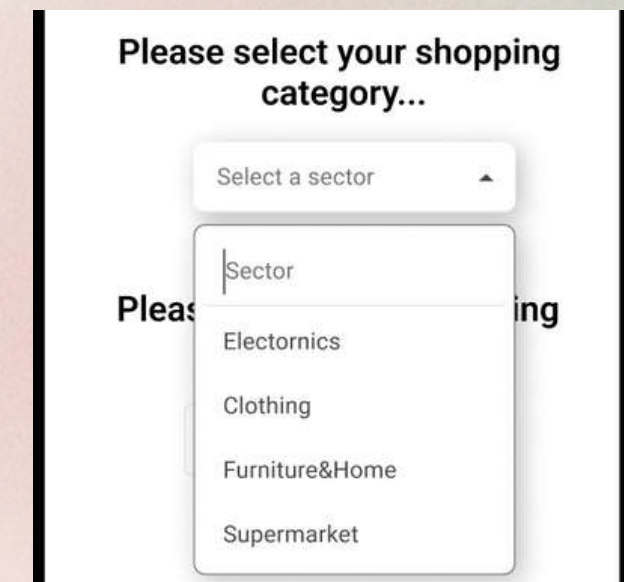
According to **Project Review** feedbacks,

Narrow Scope

We reduced our scope based on feedback and **removed one of the three** initial features to focus better.

Add Different Sectors

We expanded the app to support **more sectors** like electronics, furniture, and supermarkets, as suggested during the project review.



Reduce the Questions in Home Page

We keep only the **necessary questions**.

FEEDBACK-BASED MODIFICATIONS

According to **ADHD User's** feedbacks,

● Keep Description Short & Highlight Important Words

To improve focus for ADHD users, we **shortened product descriptions** and **emphasized key words** for better readability.



PRINTED MINI DRESS

39.95 EUR

VAT included / Exc. Shipping cost

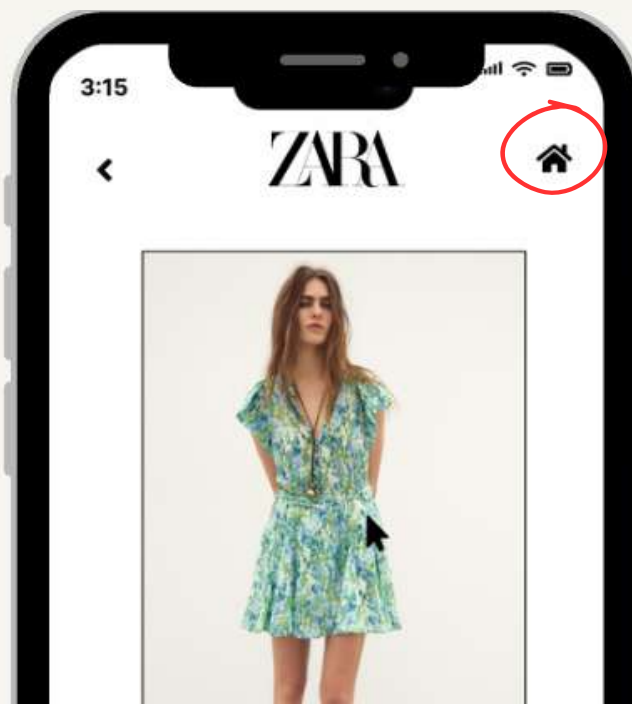
Multicoloured | 2999/258/330



A short, **flowy dress** with a **watercolor-style floral print** in shades of **blue, green, and white**. It features a **V-neckline** and **flutter sleeves**, creating a **light and breezy look**. A **thin fabric belt** cinches the waist, and the **skirt flares gently above the knees**.

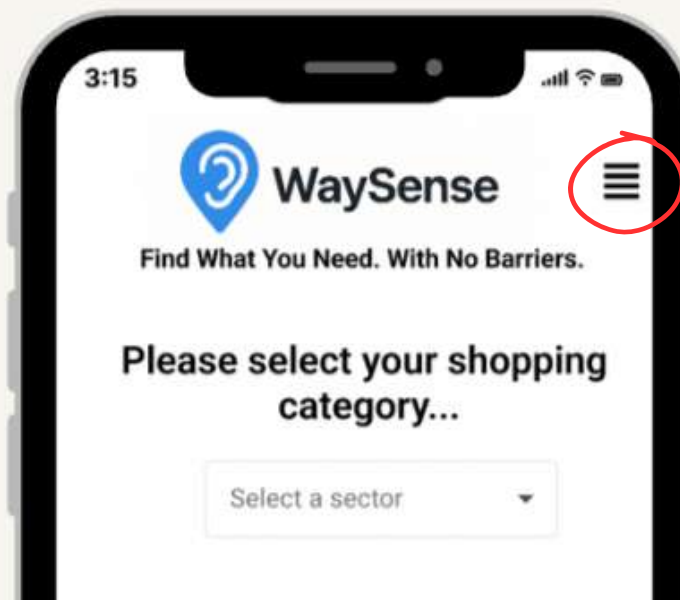
FEEDBACK-BASED MODIFICATIONS

According to **Peer-Peer Evaluation** feedbacks,



Home Page Button

We added a home button to allow users to return to the main page at any time, based on peer feedback.

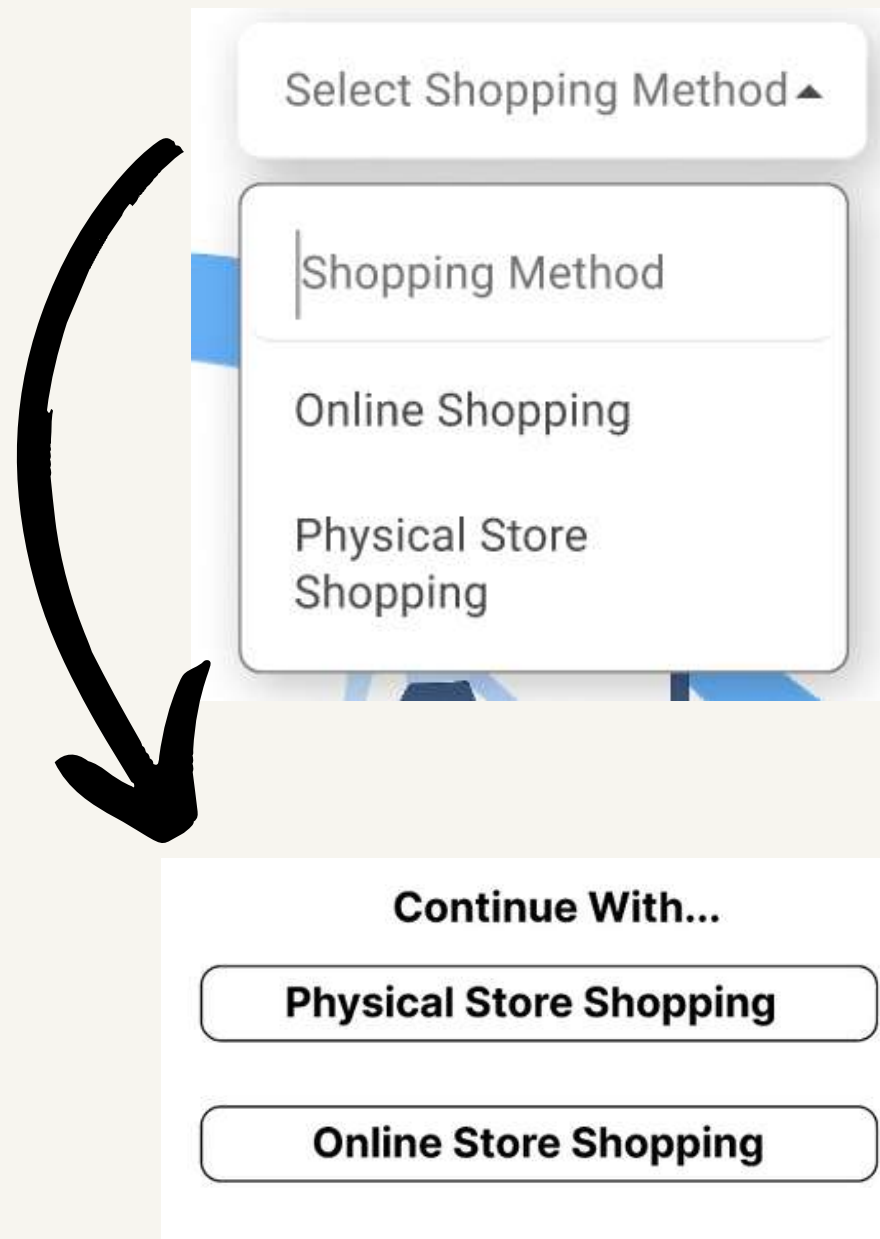


Tools Bar Button

We introduced a toolbar with quick access to Profile, Notifications, Settings, Privacy & Security, and Help—based on peer suggestions.

FEEDBACK-BASED MODIFICATIONS

According to **Peer-Peer Evaluation** feedbacks,



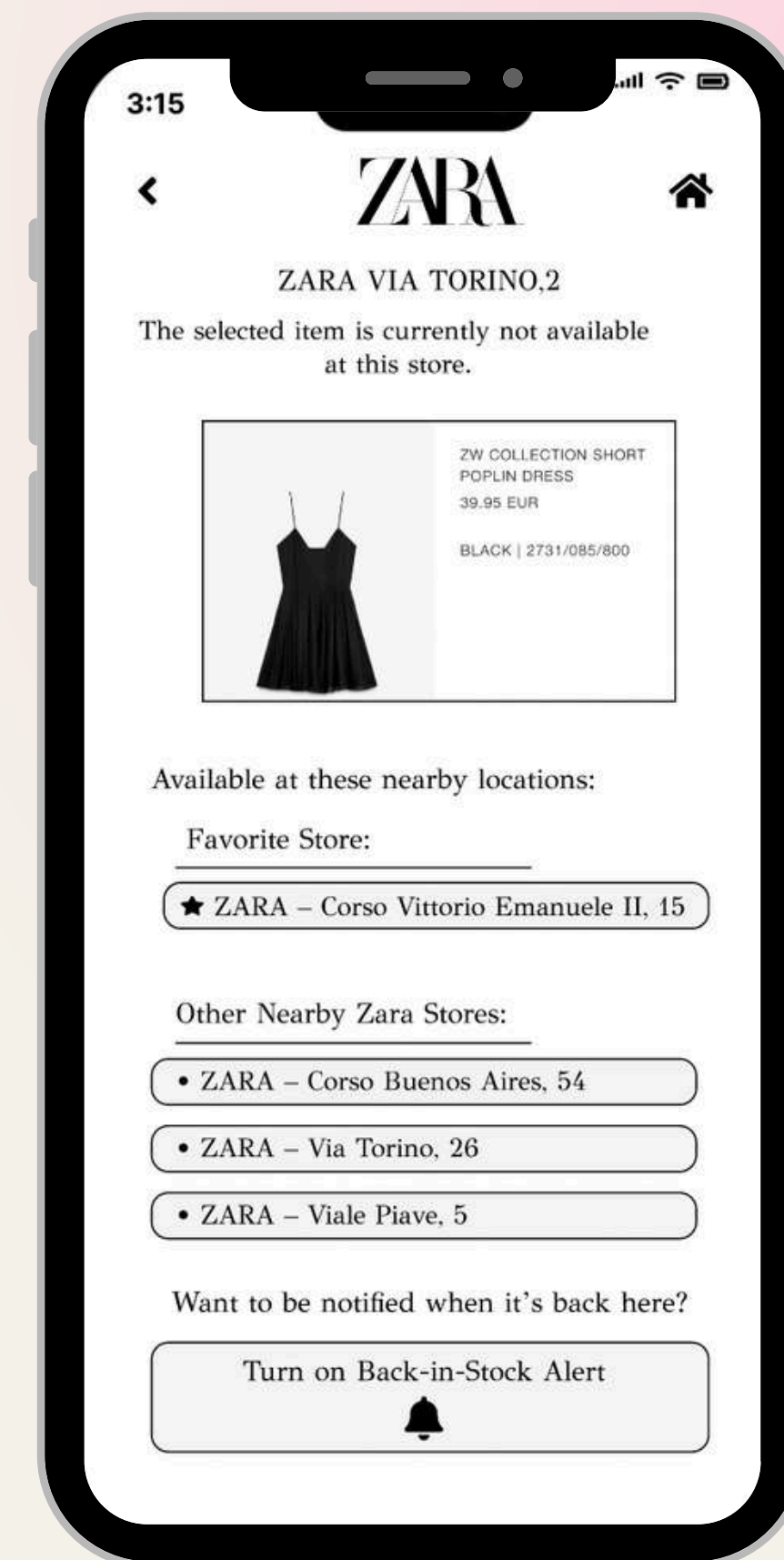
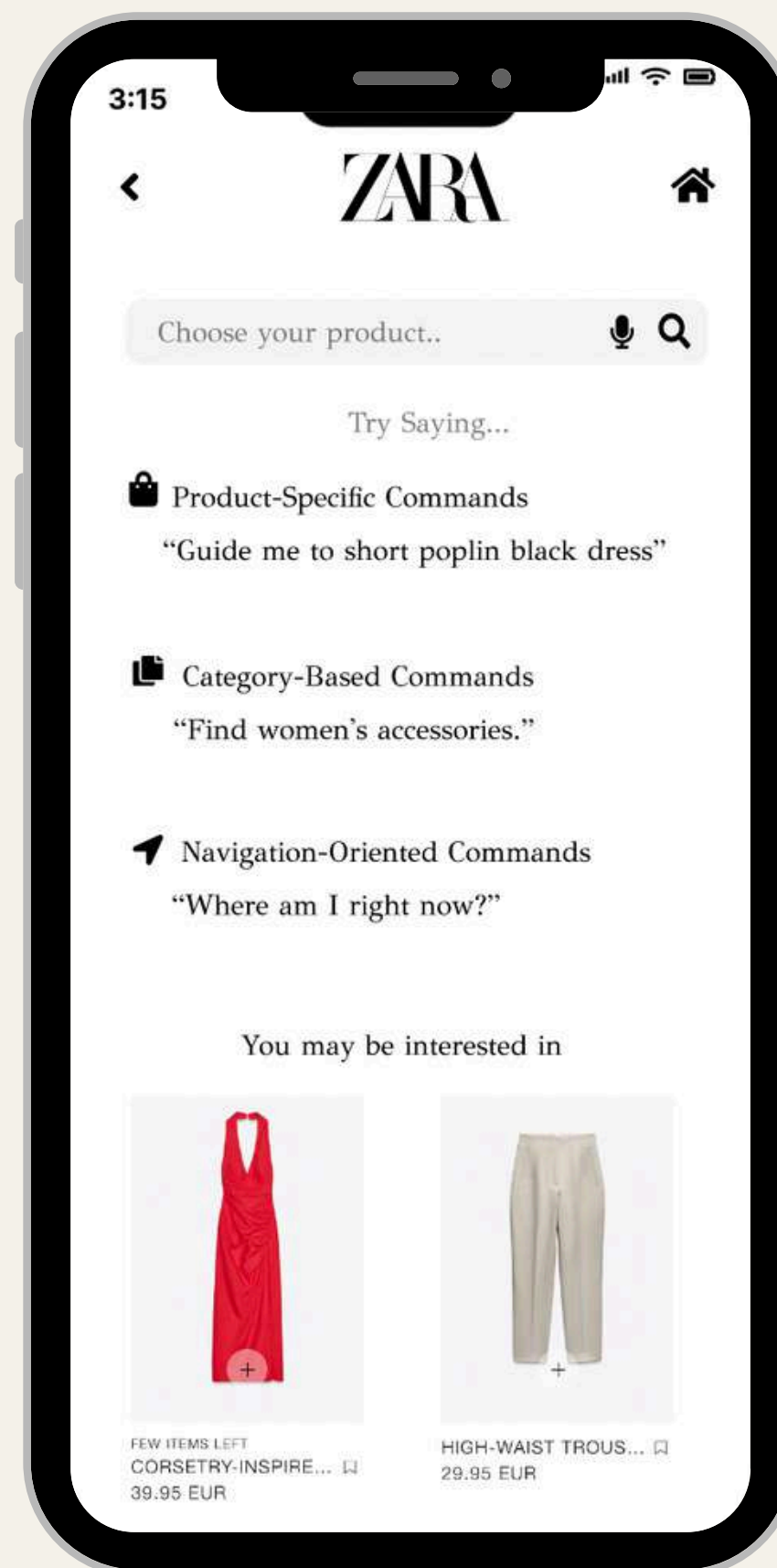
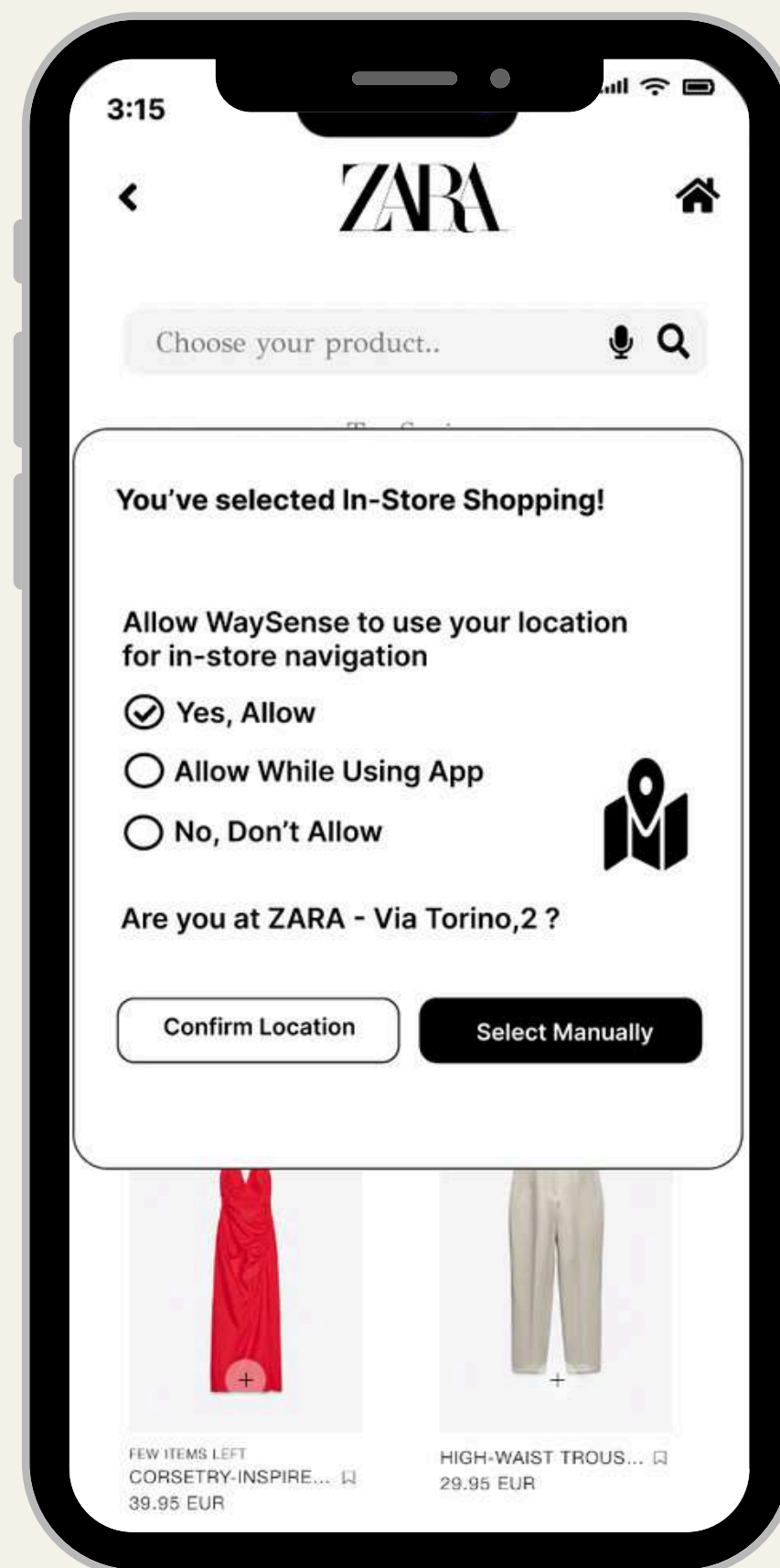
Switch Dropdown Menu to Basic Selection

We replaced the dropdown menu with two simple buttons to make the selection process quicker and more accessible.

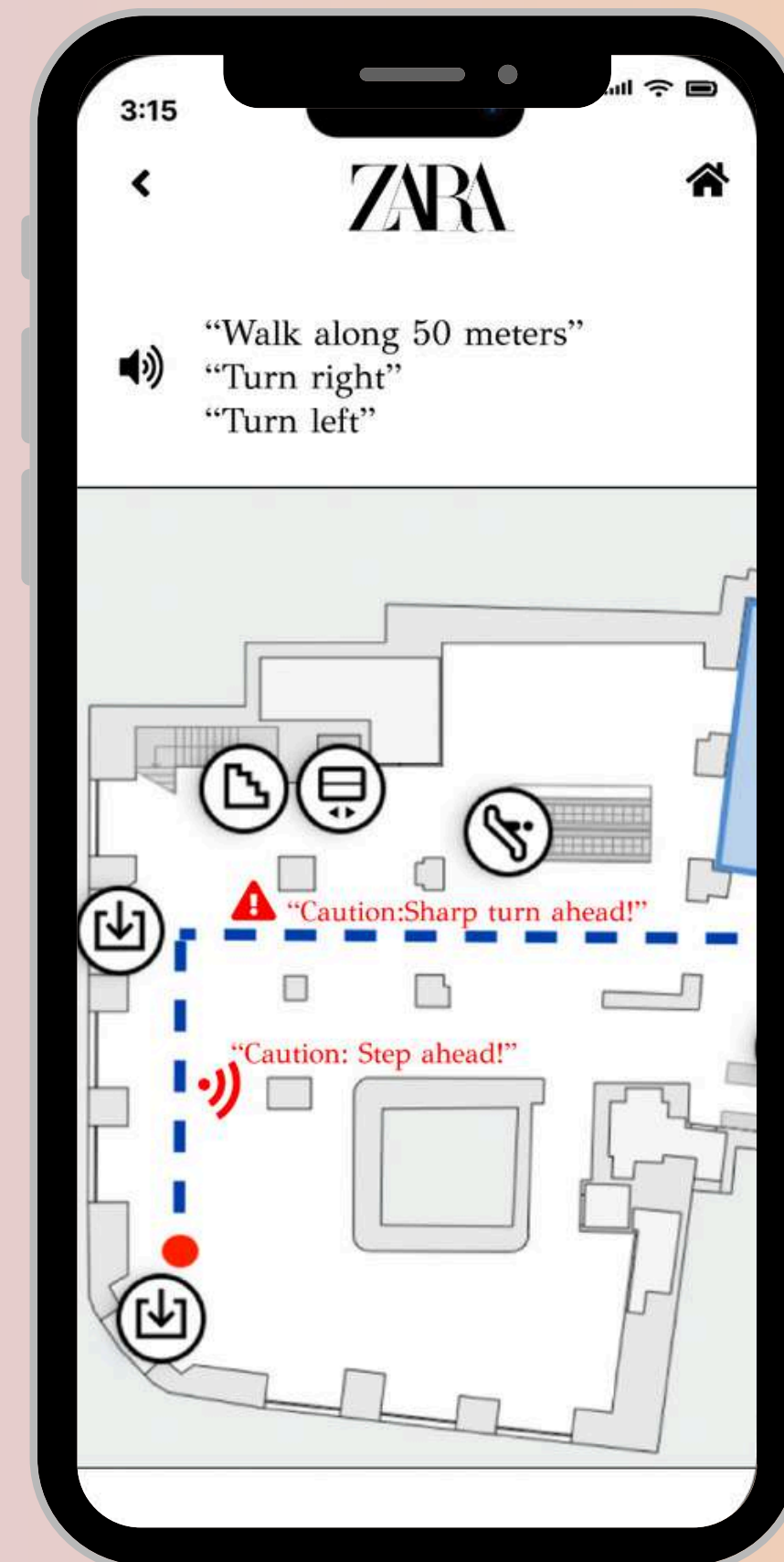
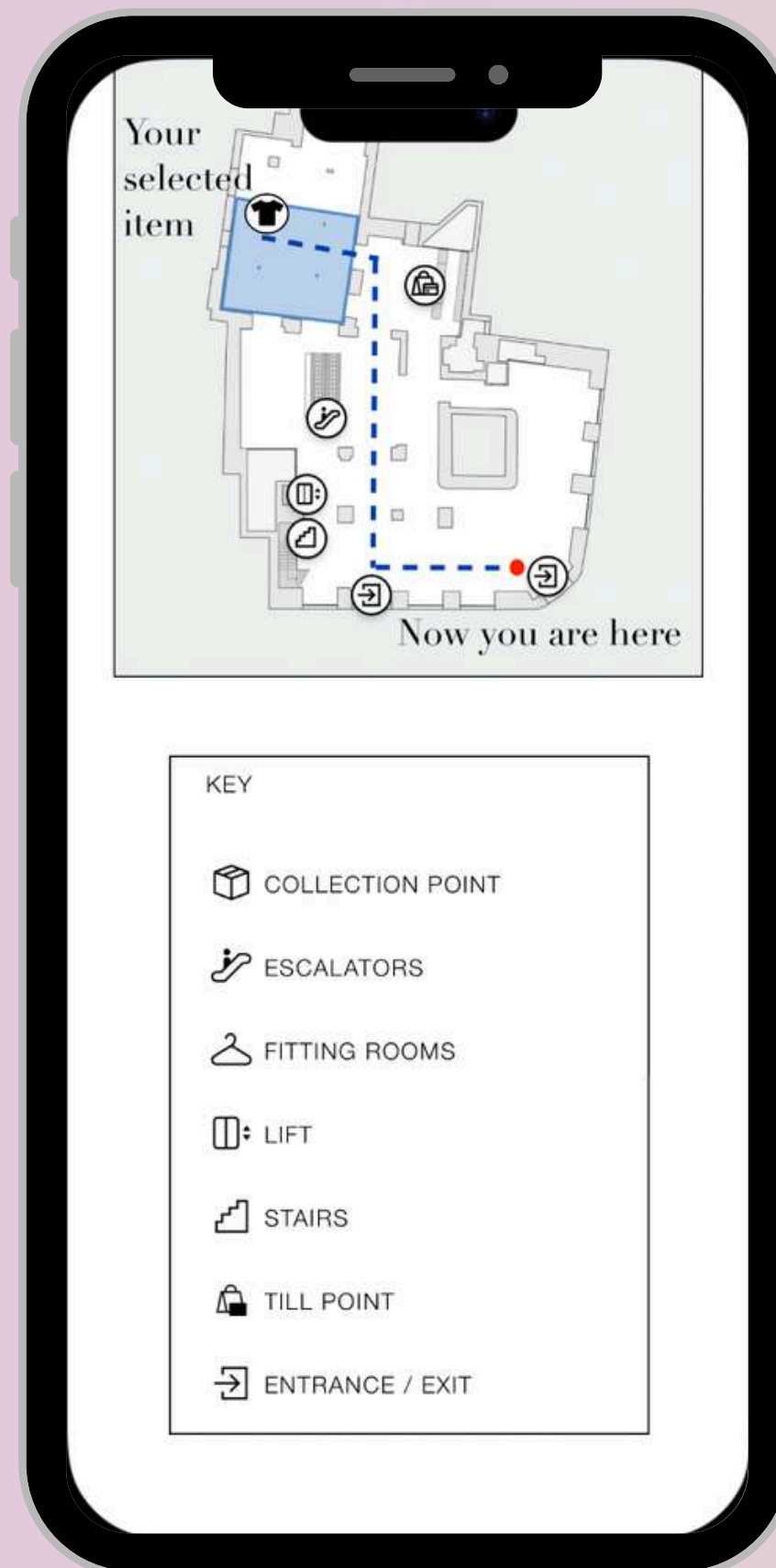
FINAL PROTOTYPE



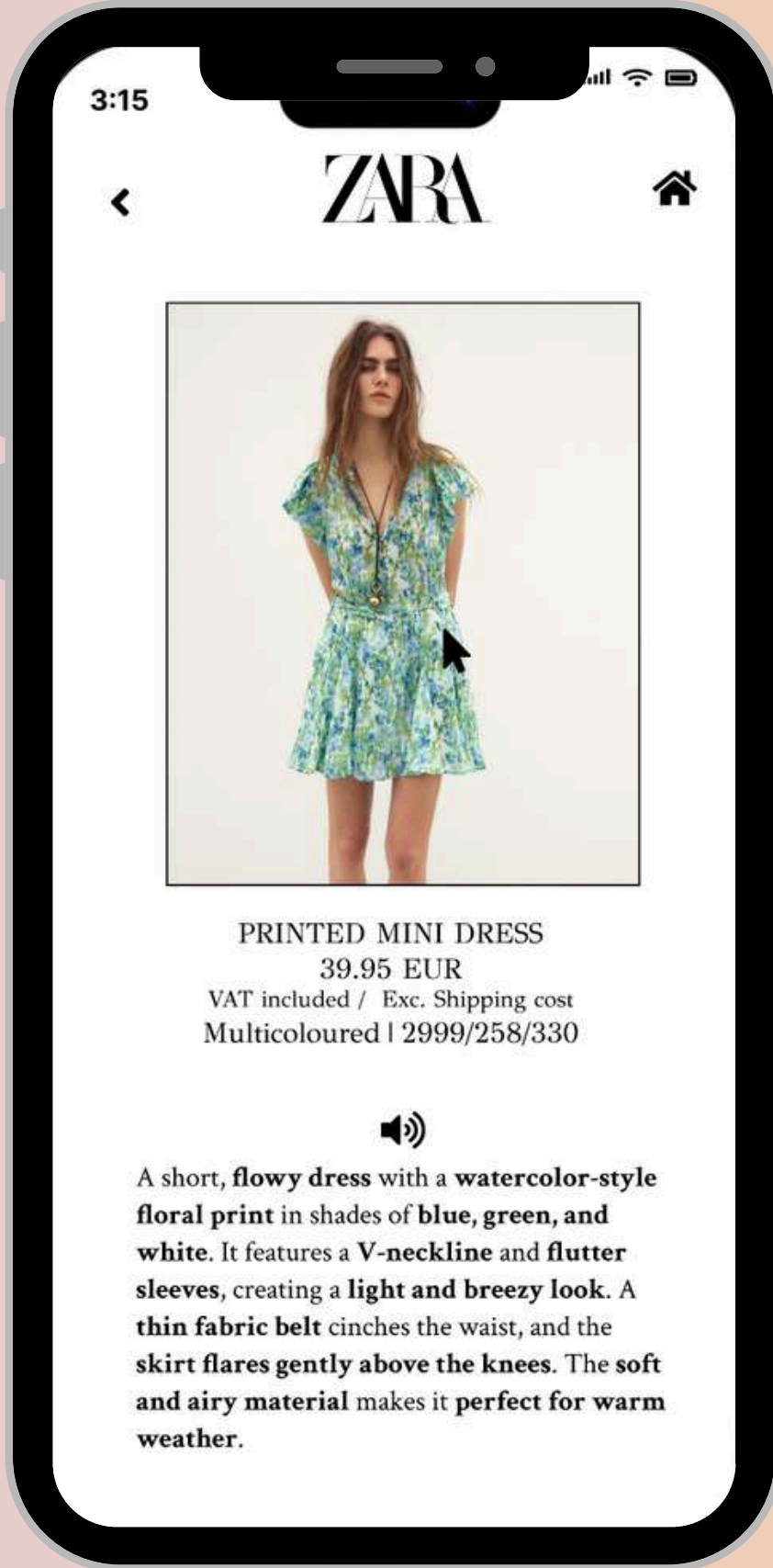
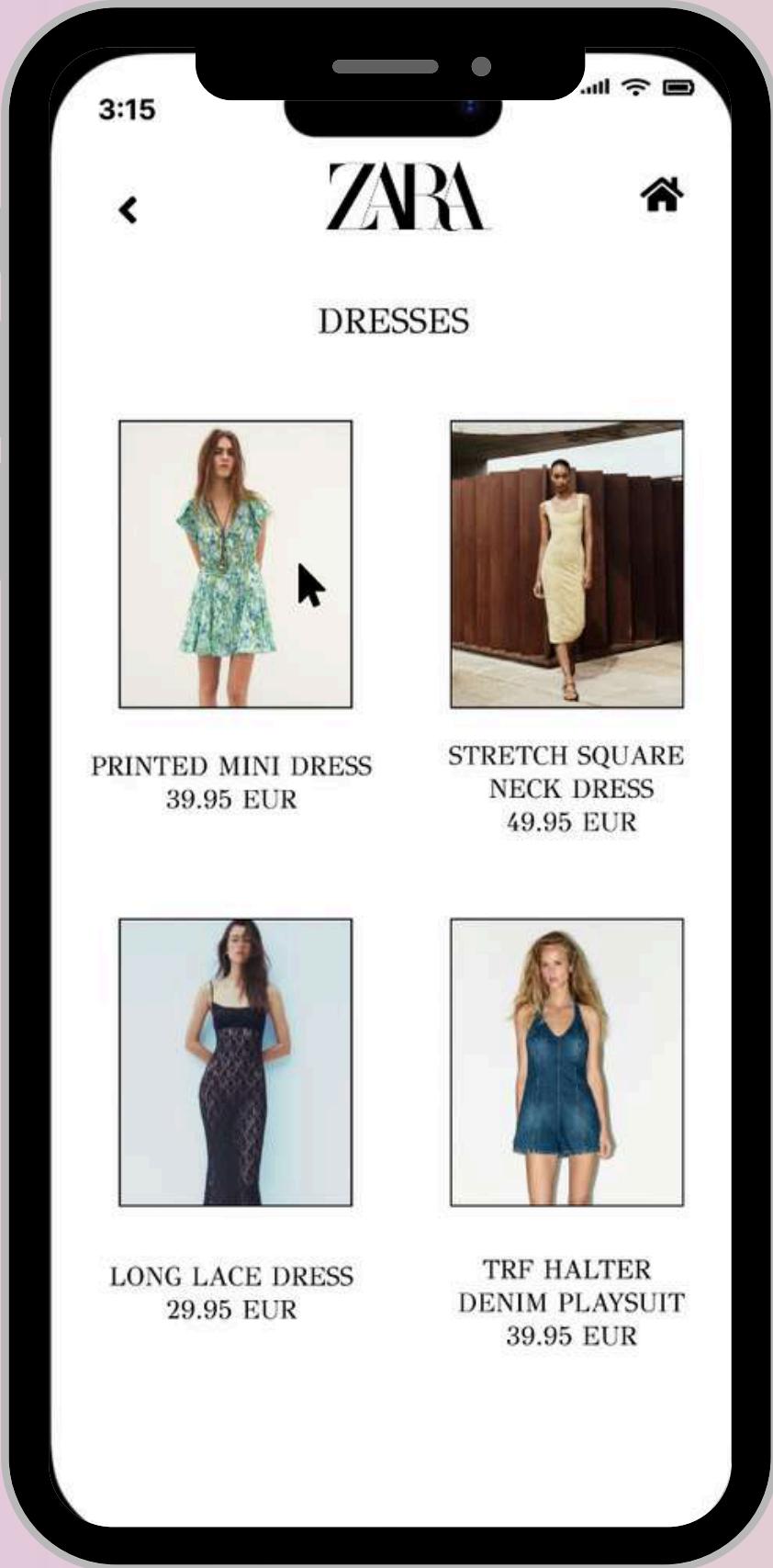
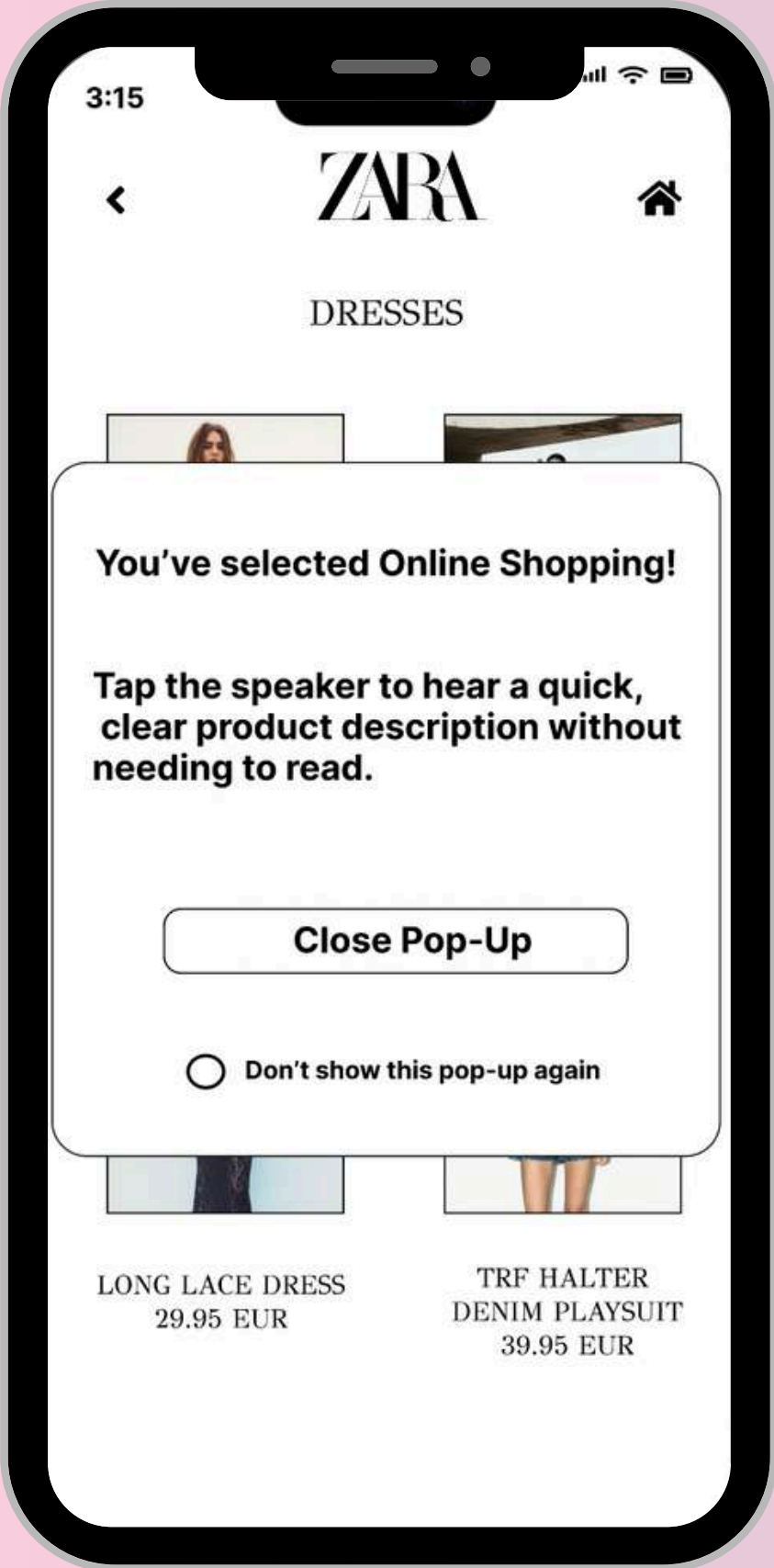
PHYSICAL STORE SHOPPING



PHYSICAL STORE SHOPPING



ONLINE STORE SHOPPING



FEASIBILITY OF PROJECT

Indoor Navigation Mapping

We developed a visual prototype to demonstrate indoor navigation. While full functionality isn't possible now **due to data limits**, the feature is **technically feasible** and could be implemented in stores in the long term.

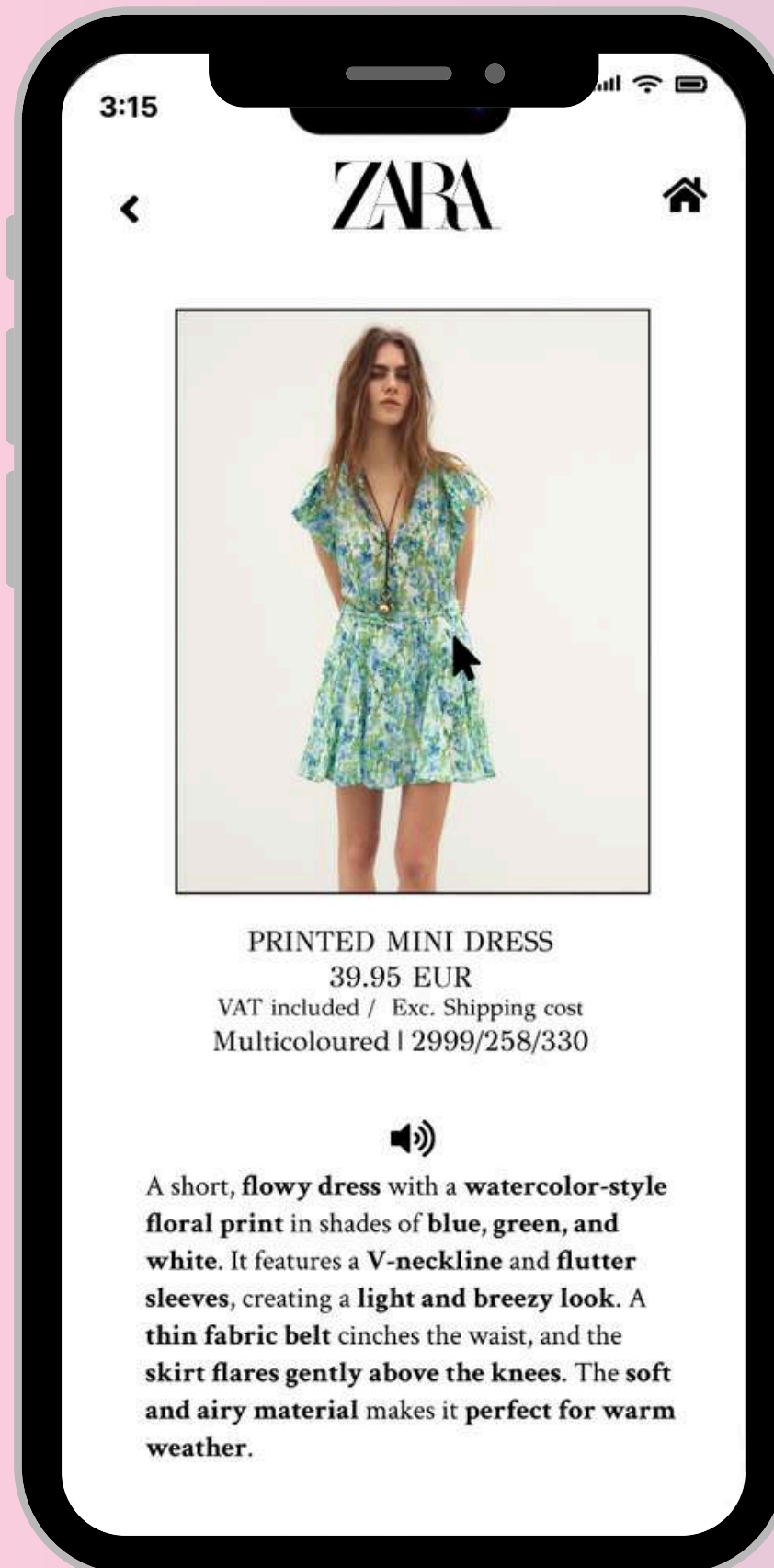
AI-Generated Product Summaries

WaySense uses AI to create short, accessible product descriptions (price, style, color, fit, fabric). Users can listen via an audio button or use screen readers. This is **fully feasible** with models like **Hugging Face LLM**, and converts it into natural speech using a **text-to-speech engine**.

ADHD-Friendly Interface Mode

WaySense uses a **minimal, low-clutter design** with step-by-step navigation and gentle focus cues to support users with ADHD. This mode was **prototyped using Figma** and **can be implemented** with basic front-end tools like CSS and JavaScript.

AI-GENERATED PRODUCT SUMMARIES



We developed a prototype to make digital shopping more accessible for individuals with ADHD. The system uses OCR to extract key product details (name, price, color, size, description), then generates a clear product summary with Hugging Face LLM, and converts it into natural speech using a text-to-speech engine. It also allows customization of voice, pitch, and speed to suit user preferences.

Our system combines OCR, LLM, and TTS to create an accessible shopping experience by extracting product details, generating clear summaries, and converting them into natural audio. This AI-driven feature can be integrated into platforms like Zara or H&M to reduce cognitive overload, making it easier for users with ADHD to focus and make decisions quickly.

USE CASE: H&M PRODUCT PAGE



The prototype scans the product interface and identifies essential visual cues and text:

These mid-length sports shorts in Sage Green are available for €29.99 and come in sizes XS to 3XL. The product page includes helpful visuals and a size guide for easy selection. Though not available in stores, shoppers can access details on fit, materials, and delivery with clearly labeled sections. The clean layout and bold “Add” button make the purchase process quick and user-friendly—especially useful for those who prefer fast, straightforward navigation without distractions. explanation of features, benefits, and usage is essential to inform and persuade customers to make a confident purchase.

DEMO

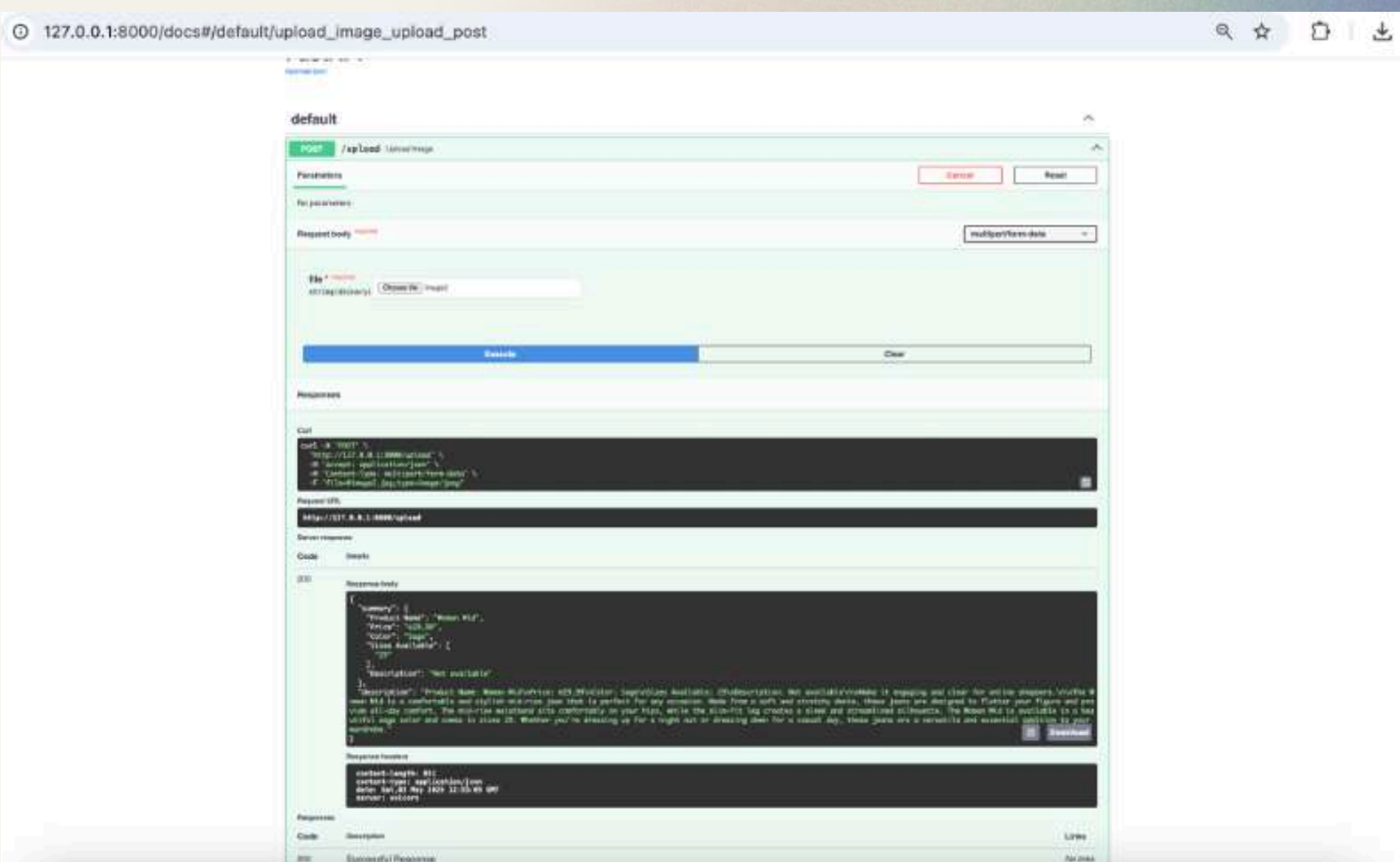
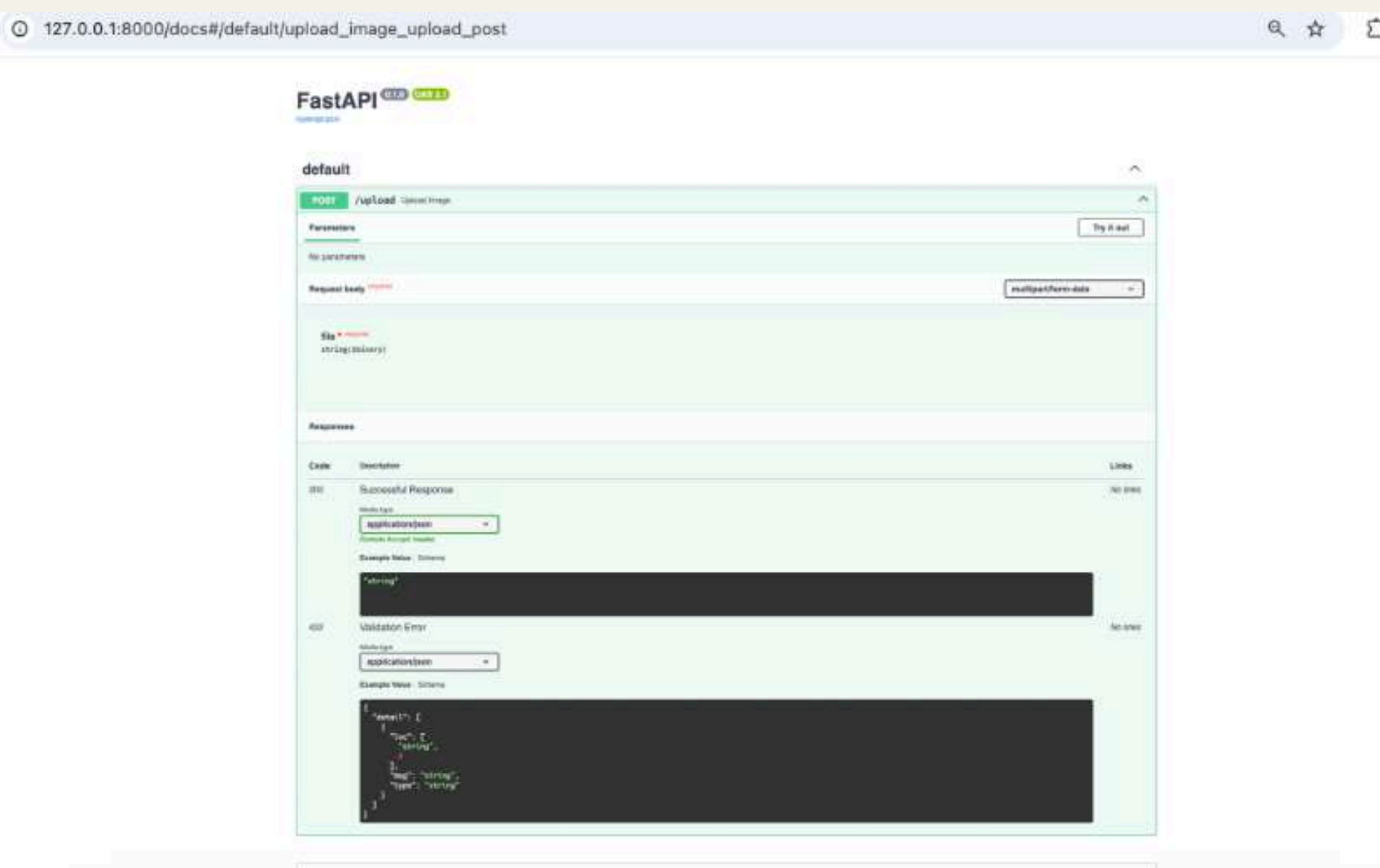
(<https://drive.google.com/file/d/18e9h-aCX6ZteaDbJySOIH7e5wXISuKUG/view?usp=sharing>).

Git Repository

- Access the Git repository at the following link: [Git Repository](https://github.com/viveksharma12527/ai_prod_voice)

AI-GENERATED PRODUCT SUMMARIES

We use FastAPI to expose a '/upload' endpoint where images can be posted. The API accepts image files and returns JSON containing product metadata and AI-generated summaries. Here is an example of an execution result with metadata and product description:



INDOOR NAVIGATION MAPPING

Every store may have different layouts depending on location, size, and country.



Need floor plans for each store (from HQ or property owners).



Maps need to be regularly updated monthly or seasonally. (if the layout changes)



Need inventory status (in stock/out of stock) is available via online store APIs

INDOOR NAVIGATION MAPPING

- Divide the store into zones and place visual markers (QR, ArUco, AprilTag) on racks.
- Use cameras and OpenCV to detect each marker's ID and location in real time.
- Link each marker to a specific zone and match products to their marker ID.
- Integrate with the store's IMS to fetch product info and stock levels.
- Generate a map showing current product availability and exact in-store location.



(a)



(b)

FUTURE WORK

Integrate **Augmented Reality** to enhance in-store navigation.

Users will navigate Zara stores in real time using their smartphone camera, with AR overlays showing:

- Directional arrows
- Step-by-step guidance
- Safety alerts (e.g., “Step ahead!”)

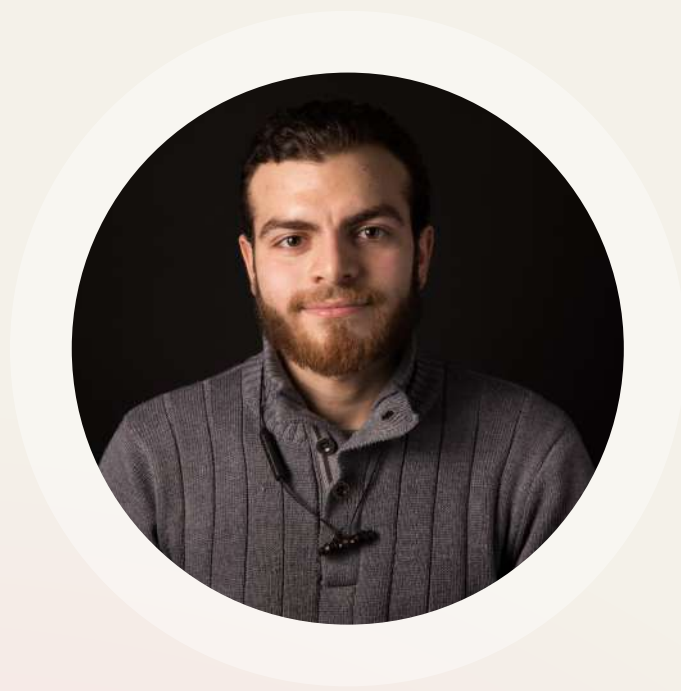
The AR view will be synchronized with the 2D store map for clarity.



THANK YOU FOR LISTENING!

IDENTIFIED BARRIERS & PERSONAS

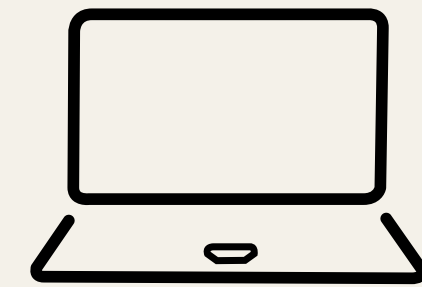
- Previous presentations we can use this informations



MARCO VITALE

Remote front end developer

33 Years Old Male Computer Engineer



Accessibility Barriers

- Struggles with focus and decision-making in cluttered interfaces
- Overwhelmed by dense grids, complex filters, and long checkout flows
- Frequently abandons purchases due to distractions or fatigue
- Suffers cognitive overload without clear guidance
- Needs structured layouts, voice/text input, and simplified navigation

Interests

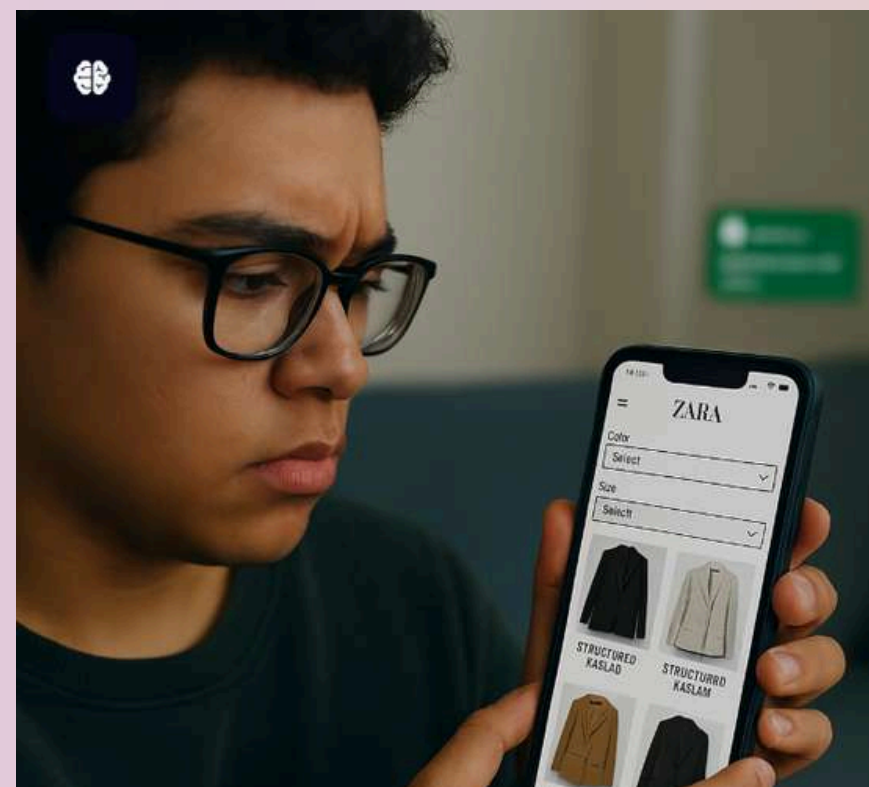
- Designing mobile apps and prototyping accessibility-first interfaces
- Playing fast-paced indie games and engaging with online ADHD communities
- Watching K-dramas and reality cooking shows while online shopping
- Using voice assistants to streamline everyday tasks

PROBLEM

Marco was browsing Zara's website to find a blazer for an upcoming portfolio review. He planned to use the filters to narrow by color and size — but the dropdown menus were inconsistent and reset each time the page refreshed.

Without voice search or preview summaries, they had to scroll through dozens of similar-looking items with vague names like “Structured Blazer.”

Overwhelmed and losing focus, Marco got distracted by a text notification and never returned — forgetting about the blazer until the event had passed.



SOLUTION

Accessibility Enhancement Toolkit

Enhance your website or app with our accessibility features for inclusive shopping.

Get Started



Voice Assistant



AI Descriptions



Screen Reader Support



Indoor Navigation

This toolkit improves the accessibility of any shopping website or app with proper contrast ratios.

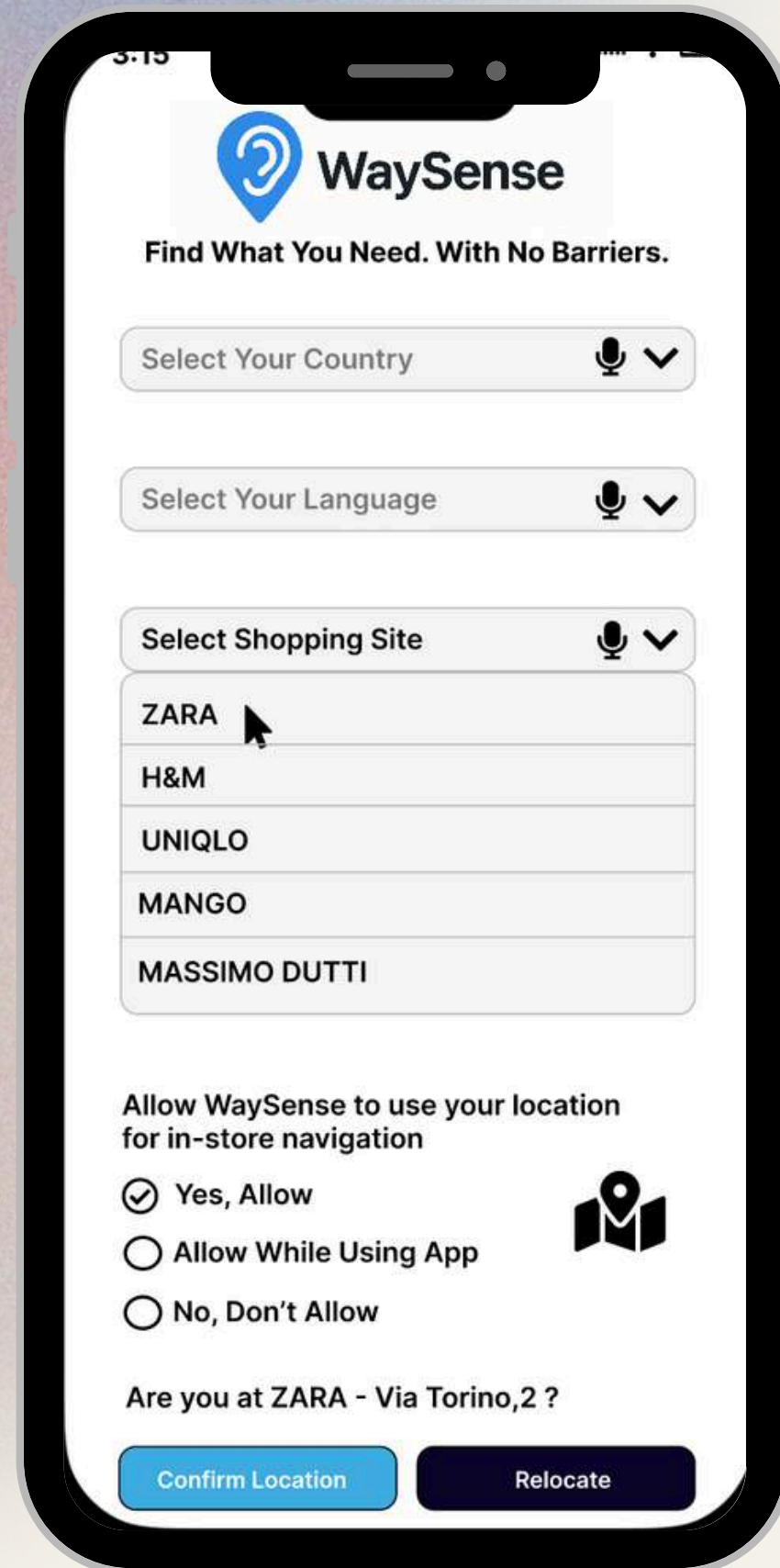
It includes four main features:

- Voice Assistant to guide users through their journey
- AI Descriptions for better product understanding
- Screen Reader Support for visually impaired users
- Indoor Navigation to help locate products in physical stores

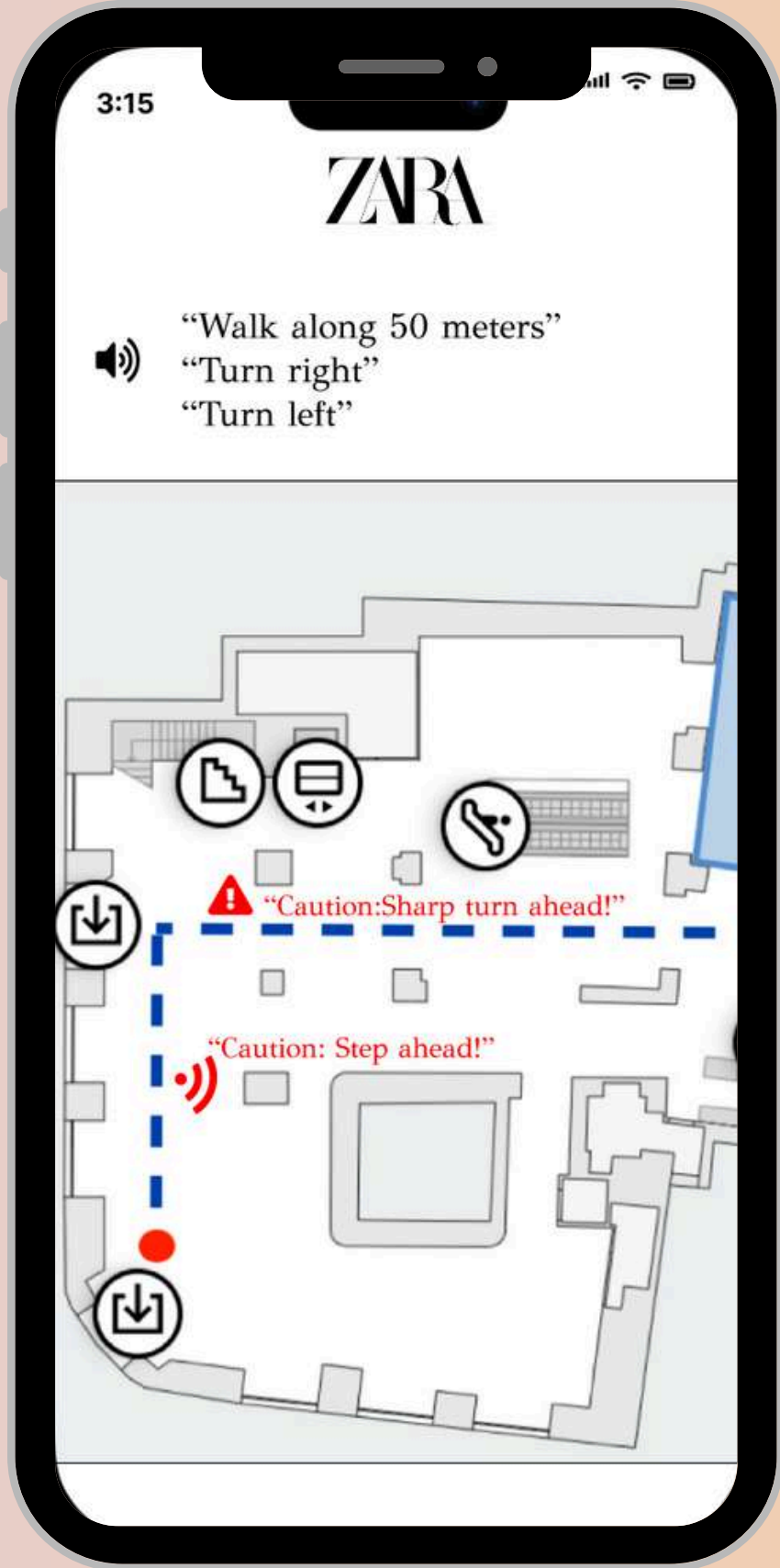
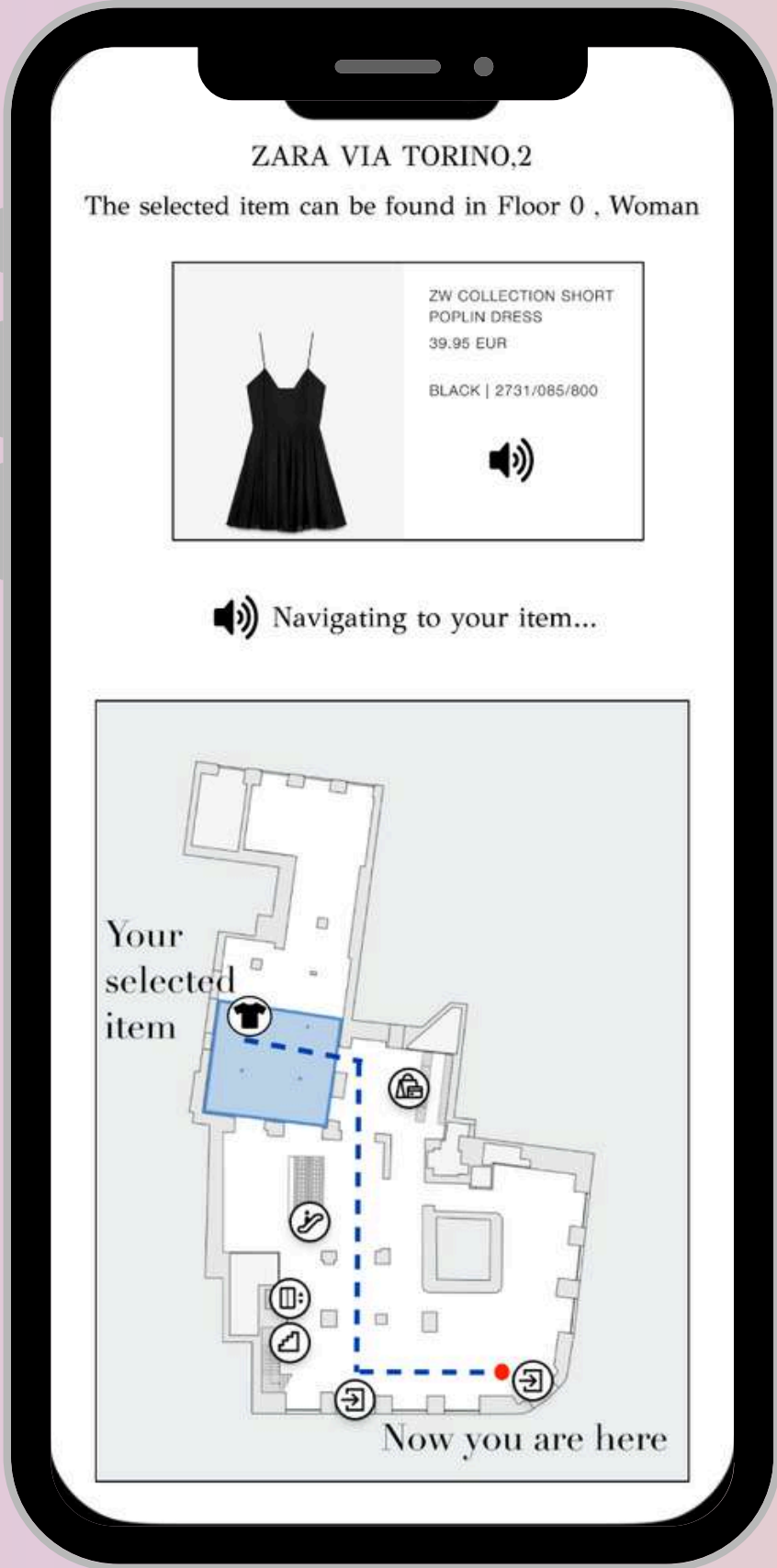
WAYSENSE

WaySense provides a simple and accessible starting point for all users.

- On this entry screen, users can choose their country, language, and preferred shopping store either by touch or voice.
- It also asks for location permission to enable indoor navigation, helping users find products with real-time voice guidance.
- This setup ensures that people with different accessibility needs can begin their shopping experience smoothly and independently.



INDOOR NAVIGATION



WHY WE FOCUSED ON ADHD

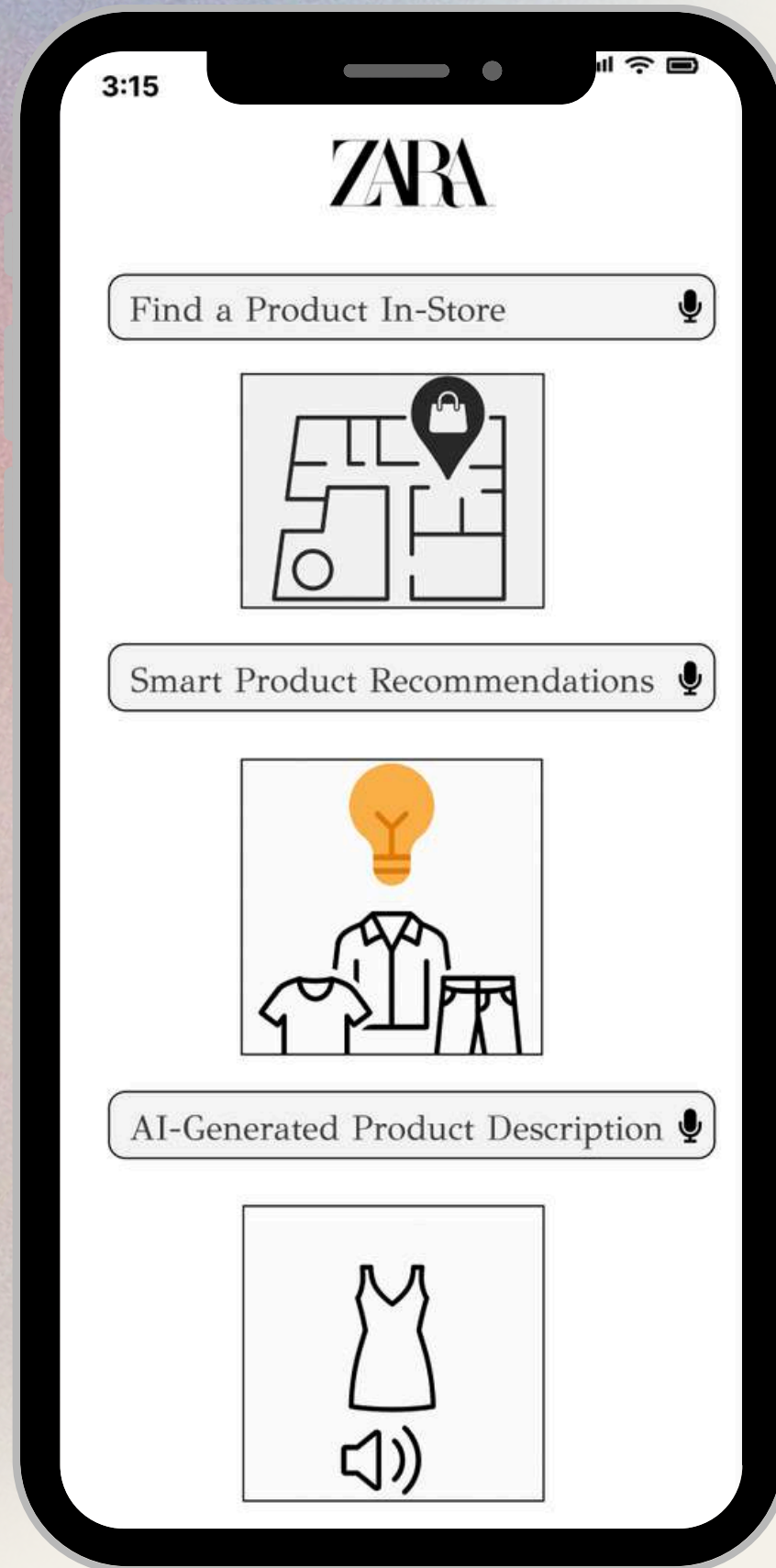
ADHD affects nearly 5% of the global population, making it one of the most prevalent neurodevelopmental disorders worldwide. Our project aims to address key challenges faced by people with ADHD, such as distraction, decision fatigue, and cognitive overload, by providing a structured and accessible shopping experience.

Condition	Global Prevalence
ADHD	~5% (general population)
Blindness	~0.5%

What We Newly Add for ADHD Users

- Clean and Minimal Interface: Reduces clutter to help maintain focus.
- AI-Powered Product Suggestions: Limits results to 2–3 tailored options to avoid overwhelm.
- Step-by-Step Navigation: Guides users through each shopping phase with simple instructions.
- Voice Command & Feedback: Enables product search and navigation via voice to simplify interaction.
- Smart Reminders: Alerts users to continue if they abandon the shopping process.

WAYSENSE

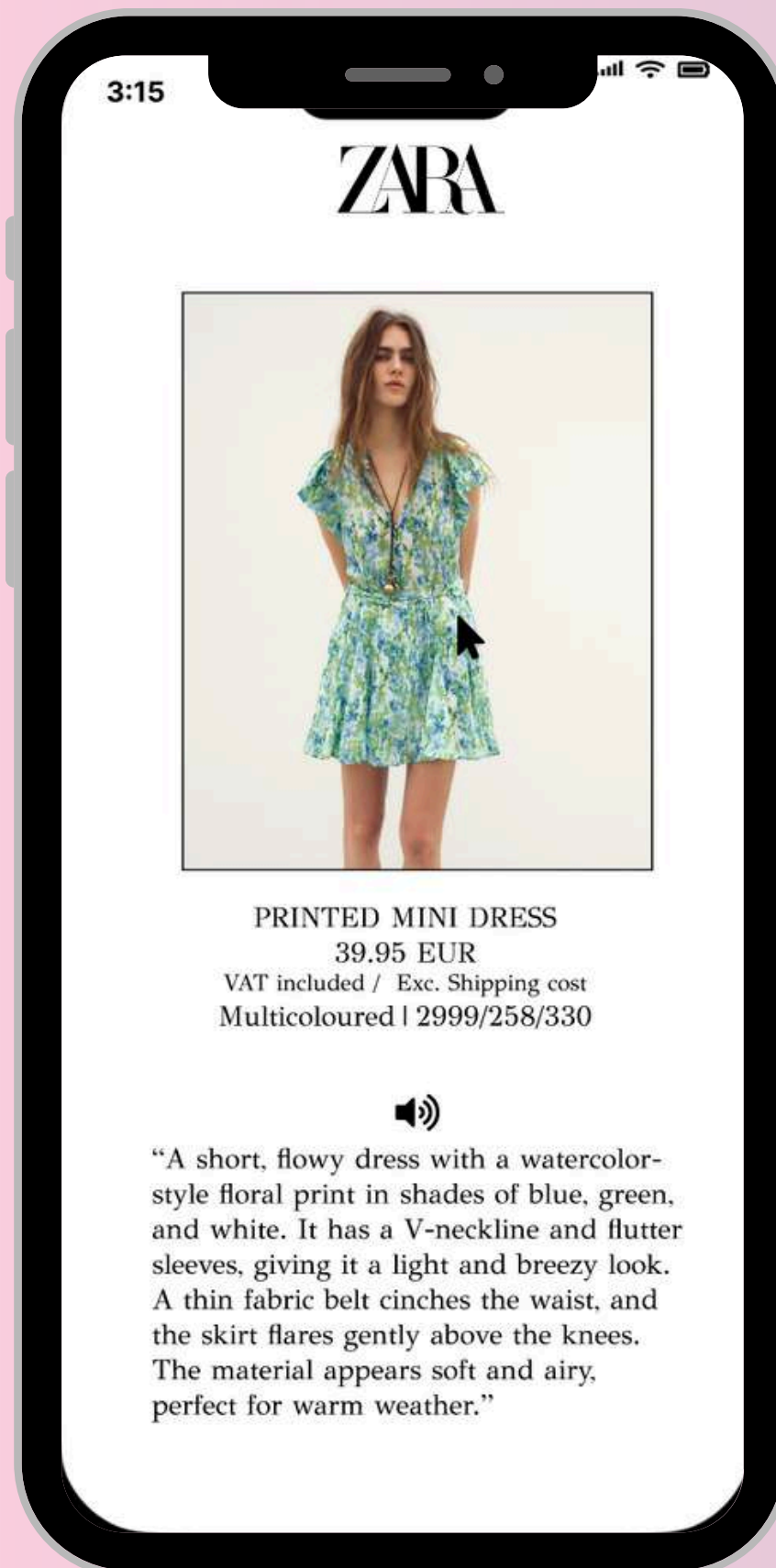


At the start of the app, users can choose between three key features:

- Find a Product In-Store – Locate a specific item on the store map.
- Smart Product Recommendations – Get 2–3 highly relevant product suggestions.
- AI-Generated Product Description – Listen to clear, accessible product details.

This simple menu helps users quickly access the feature they need, reducing overload and making navigation easy.

TOOLS FOR AI DESCRIPTION PRODUCTS

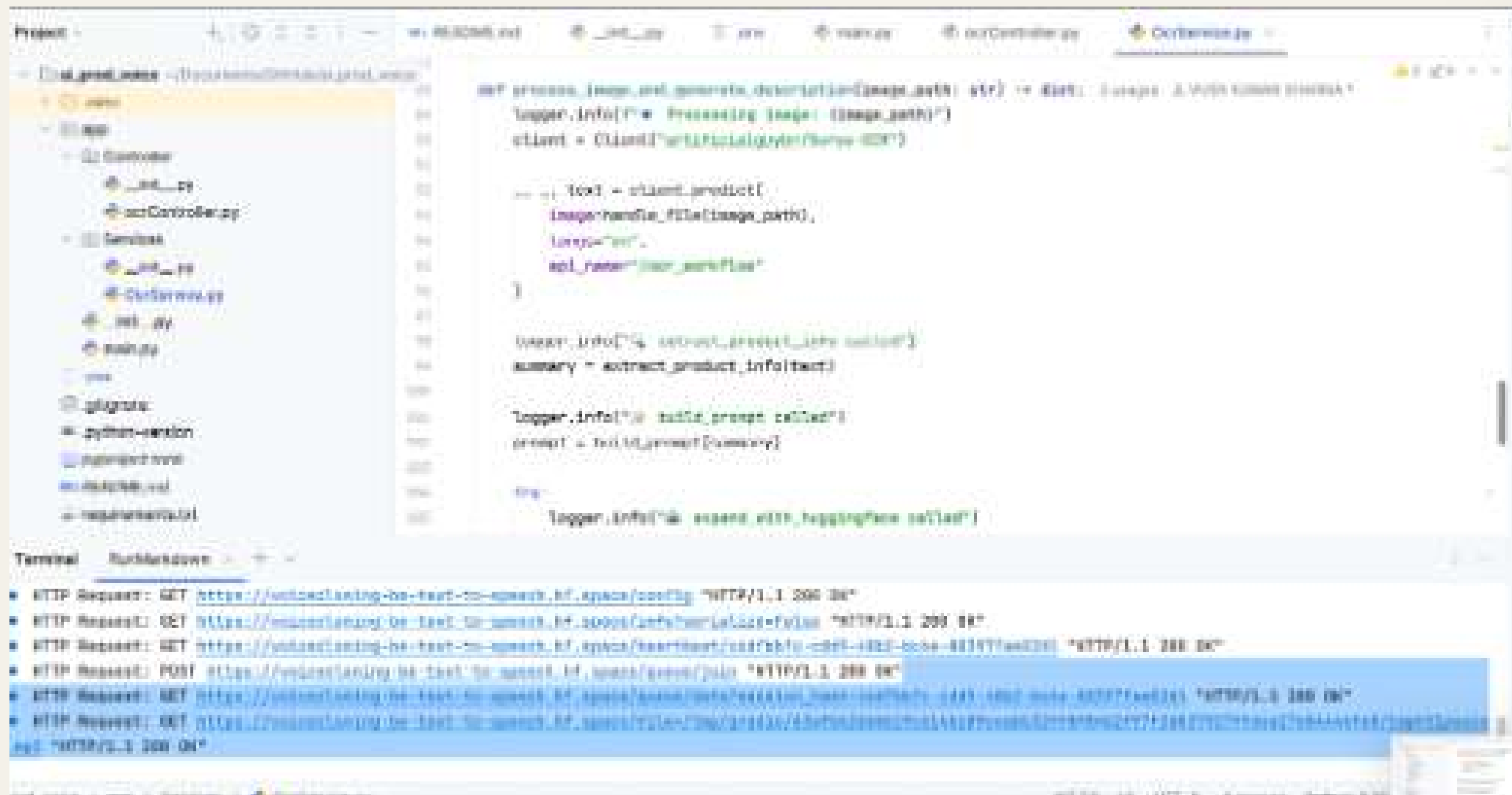


We developed a prototype to make digital shopping more accessible for individuals with ADHD. The system uses OCR to extract key product details (name, price, color, size, description), then generates a clear product summary with Hugging Face LLM, and converts it into natural speech using a text-to-speech engine. It also allows customization of voice, pitch, and speed to suit user preferences.

Our system combines OCR, LLM, and TTS to create an accessible shopping experience by extracting product details, generating clear summaries, and converting them into natural audio. This AI-driven feature can be integrated into platforms like Zara or H&M to reduce cognitive overload, making it easier for users with ADHD to focus and make decisions quickly.

TOOLS FOR AI DESCRIPTION PRODUCTS

ZARA UI ACCESSIBILITY SIMULATION



```
def process_image_and_generate_description(image_path: str) -> str:
    """Process image and generate description"""
    logger.info(f"Processing image: {image_path}")
    client = Client("https://api.openai.com/v1/gpt-4")

    text = client.product(
        image_path=image_path,
        length="short",
        style="casual"
    )

    logger.info(f"Extract product info: {text}")
    summary = extract_product_info(text)

    logger.info(f"Full prompt called")
    prompt = text + prompt(summary)

    try:
        logger.info(f"Send API request")
```

Terminal

- HTTP Request: GET <https://localhost:8080/api/openai/config> "HTTP/1.1 200 OK"
- HTTP Request: GET <https://localhost:8080/api/openai/v1/gpt-4/image/1234567890> "HTTP/1.1 200 OK"
- HTTP Request: GET <https://localhost:8080/api/openai/v1/gpt-4/image/1234567890> "HTTP/1.1 200 OK"
- HTTP Request: POST <https://localhost:8080/api/openai/v1/gpt-4/image/1234567890> "HTTP/1.1 200 OK"
- HTTP Request: GET <https://localhost:8080/api/openai/v1/gpt-4/image/1234567890> "HTTP/1.1 200 OK"
- HTTP Request: GET <https://localhost:8080/api/openai/v1/gpt-4/image/1234567890> "HTTP/1.1 200 OK"

“A short, flowy dress with a watercolor-style floral print in shades of blue, green, and white. It has a V-neckline and flutter sleeves, giving it a light and breezy look. A thin fabric belt cinches the waist, and the skirt flares gently above the knees. The material appears soft and airy, perfect for warm weather.”

HOW TO MAP STORES?

Every store may have different layouts depending on location, size, and country.



Need floor plans for each store (from HQ or property owners).



Maps need to be regularly updated monthly or seasonally. (if the layout changes)



Need inventory status (in stock/out of stock) is available via online store APIs



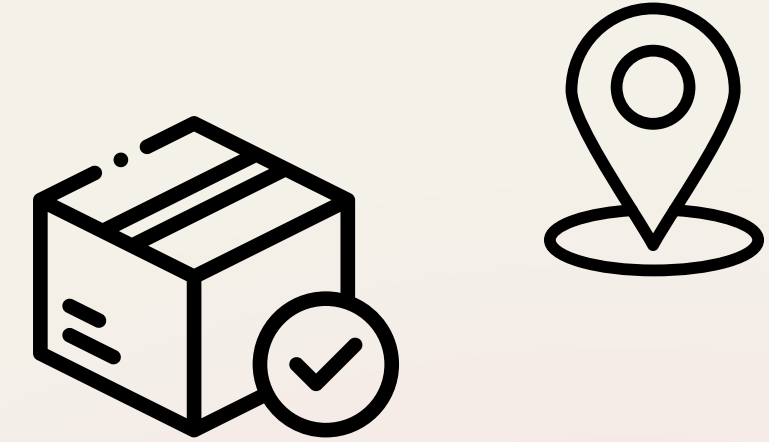
Need users to confirm their location inside the store.

TOOLS FOR INDOOR MAPPING

- Use computer vision (OpenCV + ArUco/AprilTag) to detect each marker's unique ID and position in real time.
- Build a digital map where each marker's coordinates are linked to a store section (e.g., Marker A1 = Women's Blazer Zone).
- Track marker positions continuously, keeping the store layout updated on the map.



TOOLS FOR INDOOR MAPPING



- Link products to marker IDs in the database.
- Integrate Zara's Inventory Management System (IMS) with our marker-based mapping system.
- The IMS provides detailed product data: barcode, name, category, and stock levels.
- The IMS typically shows only the general store section (e.g., Women's Blazers).
- Our system links each product to a specific marker ID representing the exact rack or shelf location.
- This integration allows the app to combine real-time stock data from Zara's IMS with precise in-store navigation. Users can see both availability and the exact product location on the store map.

REAL WORLD EXAMPLES

- Amazon's Warehouse Automation with Fiducial Markers
- MIT's Indoor Navigation Using AprilTags



(a)



(b)

FEASIBILITY

AI-Generated Product Summaries

WaySense will integrate AI-generated summaries for each item (e.g., fabric, fit, color, style). These are available via an audio button on the product page and also compatible with screen reader tools. Each product has a short, accessible description focusing on: style, color, fit, fabric and ideal occasions or pairings.

This is fully feasible using GPT-4 or similar models. Product metadata (image, name, tags) can be passed to the model. Descriptions can be rendered as ARIA-labeled elements or embedded audio.

ADHD-Friendly Interface Mode

- Reduced visual clutter
- Step-by-step shopping (first select type, then preferences, then results)
- Visual timers or “focus mode” nudges

Requires basic front-end development (CSS toggles, JavaScript state management) and can be prototyped easily.

Voice-Based Guided Shopping

Users on Zara’s site will access a voice-enabled search bar. This supports ADHD users who prefer auditory control over complex menu structures. Users can use voice commands such as:

“Show me jackets for spring under 60 euros”

“Find comfortable outfits for travel”

Then, guided flow is triggered that it filters product options and narrows choices to 3–5 tailored recommendations.

This is technically feasible using the Web Speech API for speech-to-text and

text-to-speech, paired with Dialogflow or another NLU (natural language understanding) engine. Can be easily implemented within the browser environment without backend dependency.

FEASIBILITY

Indoor Navigation Mapping

We are currently developing a prototype, but at this stage, it is not feasible to implement full mapping with QR codes/AprilTags, access store cameras, build a detailed store map, or obtain stock data beyond what is available in the Zara app. However, based on the tools we have researched, these features could become implementable in the future once the necessary data (such as store layouts and product placements) is accessible. While we cannot yet develop a fully functional version, it is feasible to create a visual prototype that demonstrates how the system would work.

-  UI and static map preview are feasible.
-  Accurate store layouts & product placements require a partnership with Zara.
-  QR code / AprilTag mapping is feasible and avoids BLE/Wi-Fi.
-  Only section-level guidance is possible; exact shelf locations aren't available.
-  Users must scan markers manually; no continuous tracking.

FUTURE IMPROVEMENTS

- Plan to integrate Augmented Reality (AR) indoor navigation for a more immersive experience.
- Users will navigate Zara stores in real time using their smartphone camera.
- AR overlays will display:
 - Directional arrows.
 - Step-by-step guidance.
 - Safety alerts (e.g., “Step ahead!”).
- The AR view will be synchronized with the 2D store map for clarity and accuracy.
- This feature will provide clear, direct visual cues, enhancing accessibility—especially for users with ADHD.
- Aims to make in-store navigation faster, easier, and more intuitive.

